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SENIOR PROJECT (II)

Smart Speaking Signs Device

2024-2025

Students Statement

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ABSTRACT

With the progress made in technology over the 21st century, helping aids have appeared to make things easier to access for those who are impaired. The mute and deaf suffer most from being completely cut off from communication with people who have no concept of sign language, hence social as well as employment handicaps.

Research indicates that millions of individuals use sign language as the main communication tool. Yet, due to the inability to have a common perception towards sign language, they are unable to communicate among themselves in most areas of everyday life. Current solutions have been implementing partial translation or basic gesture recognition, but are not able to devise an overall effective and efficient communication aid.

Smart Speaking Signs Device (SSSD) resolves this challenge by enabling real-time sign language user translation into text and also verbally. The Smart Speaking Signs Device (SSSD), which enables real-time sign language user translation both vocally and into text, provides a solution to this issue. The system enables the harmonious coexistence of sign users and the general public by utilizing contemporary sensors, wireless connectivity, and support for mobility apps. The system can be easily implemented to ensure inclusion in government services, companies, and educational institutions since it is affordable and easy to use.

A functional prototype was developed using flex sensors, an IMU sensor, and an ESP32 microcontroller to capture hand gestures, which are then processed and transmitted via Bluetooth to a mobile application. The app displays the corresponding translated text and plays it through text-to-speech, enabling effective and independent communication in real-world scenarios.

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LIST OF SYMBOLS AND ABBREVIATIONS

LIST OF SYMBOLS	
\$	Dollar
M	Micro
°C	Celsius temperature
LIST OF ABBREVIATIONS	
SSSD	Smart Speaking Signs Device
IoT	Internet of Things
APIs	Application Programming Interfaces
3D	Three-Dimensional
AI	Artificial Intelligence
MPU6050	Sensor for motion tracking
I/O	Input / Output
OCR	Optical Character Recognition
TTS	Text-To-Speech
IMU	Inertial Measurement Unit
ISO	International Organization for Standardization
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
NA	Not Applicable
ML	Machine Learning
Wi-Fi	Wireless Fidelity
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission

CHAPTER 1: INTRODUCTION

Deaf and dumb people face a lot of difficulties in dealing with others, and this is one of the various reasons that keeps them from integrating into society. Most of the methods currently in use for communication depend on the use of sign language; it is an effective system among the practitioners of this language, yet not well known to the general public. This breakdown in communication often forces them to withdraw socially and isolates them further in obtaining essential services.

With this in mind, this project is directed toward the development of a new device that incorporates sensor technology with the capability of converting visible signs into audio and printed texts. This will then be used as an assisting device directly for the deaf and dumb to cater to their needs and express their ideas without necessarily having to rely on sign language experts.

This project represents a number of distinctive features provided by the ease of use, low cost, and immediate feedback not available in current technologies.

It is not just only to solve the problem of the communication barrier that is included in its mission, but it also facilitates social integration and spreads awareness in society at large about the importance of standing in support of deaf and mute individuals. This contributes to the advancement of assistive technology in general and provides innovative solutions to meet different cultural and linguistic needs. It thus aims to develop some innovative solution mechanisms to help deaf and mute people adapt to current social settings easily. This is qualitative leaping for the sake of equality in societies—a leap, one may well say—to free expression and communication among people without barriers.

1.1 Problem Statement and Purpose

Around the world deaf and mute people are facing significant communication barriers, with an estimated 70 million people expected to be affected by 2024 [1]. Although people that are aware of sign language can communicate in their community, sign language is not universal and the barrier of communication limits interaction with society as a whole. This communication barrier causes a sense of detachment and ostracism.

Many assistive technologies have been invented and developed, but most products are too cumbersome, too expensive or too complicated, as well as missing real time experience or usability. Our project will augment this field by design a wearable assistive device - a smart glove with integrated sensors - that converts hand gestures into spoken words and on-screen text via a mobile application.

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Overall, the purpose of the project is to bring deaf and mute people closer to the hearing community to communicate in certain key contexts such as hospitals, as time is of the essence, or to eliminate barriers in situations such as public transport where communication could otherwise impede the efficiency of the system, and regardless develop a quality of life that enables the users to independently express their needs.

1.2 Project and Design Objectives

The following objectives specify the scope of the project and its measurable outcomes and, in doing so, lay down concrete success criteria for evaluation:

1. **Develop a gesture recognition system:** Create and design a system with sensors to recognize gestures and then interpret them accurately.
2. **Include real-time audio conversion:** Develop a feature that converts the detected gestures into audio and then plays them through the speakers with a delay of no more than one second.
3. **Implement a text display feature:** Create a feature that displays the detected gestures as printed, readable, and formatted text on the screen with a delay of no more than one second.

1.3 Intended Outcomes and Deliverables

Intended Outcomes:

1. **Better communication:** Using hand gestures for deaf and mute people to better communicate with others when they are had to.
2. **Empowering society:** To increase the social participation of deaf and mute people to give them more education and job opportunities.
3. **Technology for humanity:** Using modern technology (IoT) for humanitarian or accessible purposes.
4. **Raising awareness:** Highlighting the issues that deaf and mute people face and how technology can assist with those issues.
5. **Empowerment and Independence:** We hope to increase the independence of deaf and mute people in everyday life by reducing their reliance on an interpreter - especially in serious situations such as at a hospital, or a public place, where communication needs to be clear and or timely.

Deliverables:

1. **Gesture-to-speech device:** A working prototype of a device equipped with sensors that can recognize pre-defined signals, translate them into sound, and display the results as text on a mobile application.
2. **Smart glove:** Equipped with the required sensors and wireless communication unit (Bluetooth) to connect it to the application.
3. **Mobile application:** User interface: Create an easy-to-use interface that displays words and texts.
4. **Experiments and tests:** Testing the glove with a group of deaf and mute users and collecting their feedback to improve the system.
5. **Illustrations or videos:** An illustrative video of using the glove and the application, specifically in hospital settings to assist deaf and mute individuals.

These results and final products ensure that the project not only achieves its technical goals but also achieves the overarching purpose of impacting society.

1.4 Summary of Report Structure

The senior design project document consists of the following chapters and is arranged to be conveniently readable by the technical people. It begins with chapter One, which introduces the project by outlining the challenges faced by deaf and mute individuals, the motivation behind the system, its objectives, and the expected deliverables. Chapter Two contains a brief description of existing systems that are partially similar to our system. Chapter Three gives a list of hardware components that will be used in our system and explain why we chose them over other options, algorithms, and flowcharts that our system uses to perform its functions. Chapter Four focuses on results and discussions, analyzing the system's performance and accuracy based on simulations and experimental tests, which will be fully completed in Senior Project II. Chapter five, covers Tasks, Schedule and Milestones of the proposed system, Resources, and cost management that our system uses to perform its functions. In chapter Six, we talk about the impact of our project economic, societal, and global, then we discuss environmental and ethical issues that must be taken into concern. Finally, chapter Seven concludes the report by summarizing the project successes and achievements, providing the skills and experience gained, and providing detailed recommendations for further future improvements. In this way, it ensures that the SSSD project presentation is well documented and technically sound.

CHAPTER 2: BACKGROUND

2.1 Overview

Thus, improvement in sensor technology and controllers have paved the way for the advent of wearable devices to interpret human gestures. With the use of flex sensors, accelerometers, and microcontrollers, such as Arduino and Raspberry Pi, devices able to detect hand and finger movement with high accuracy have come into existence. In recent years, their applications were increasingly popular in healthcare and communication technologies to support accessibility and inclusion.

This project will use smart gloves that use flex sensors attached to each finger, detecting finger movement. The signals are processed by the microcontroller and transmitted to a mobile app wirelessly, where the app shows the corresponding text on its interface and can produce voice output. This device would therefore make it easy for people with either hearing or speech impairment to express their views and needs without having to rely on interpreters.

The design of this project focuses on the provision of real-time feedback, ease of use, and cost-effectiveness. This system gives instant conversion of sign language into spoken words and text, thus easing communication and encouraging a sense of independence among its users.

2.2 Related Work

This section illustrates, several projects similar to the concept of this project will be studied, and a comparison will be drawn between them and this project. We will present the following from these projects:

1. Project to convert sign language for the deaf and mute into audible speech using cameras and artificial intelligence

This project, therefore, proposes the development of a system using cameras and artificial intelligence (AI) to convert sign language used by deaf and mute people into verbal language to bring people to talk with each other and (or integration). The system will use a camera to capture hand and finger gestures (as illustrated in Fig. 1), analyze them using AI algorithms to recognize moods and bring them to spoken language, thus helping in preventing social alienation and providing a practical medium for interactive communication. This project will be a great aid in enhancing socialization, interactive education of sign language, and so on. However, it will face challenges that

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include, among others, the possibility of its low accuracy with regard to gesture recognition, high development costs, limited access to the technology it depicts, and the adaptation period any potential customer would need to go through for the system, as well as the translators [2].

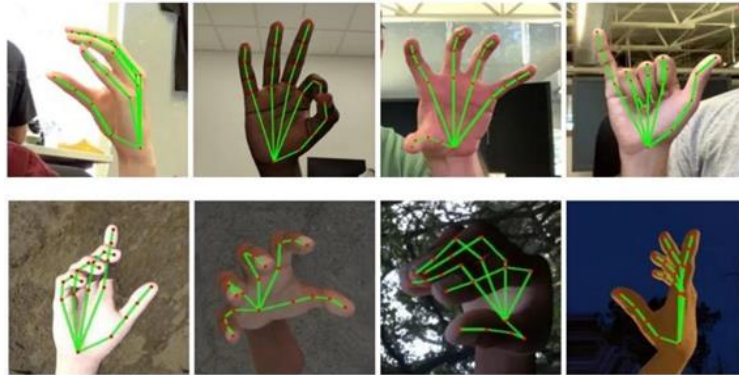


Figure 1: Convert Sign Language project [2]

2. IoT based assistive device for deaf, dumb and blind people

This project presents an Internet of Things (IoT) -based assistive device for deaf, dumb, and blind people using Raspberry Pi and Google Application Programming Interfaces (APIs) to offer three main functions. Blind people can hear the text through images, using optical character recognition (OCR) and text-to-speech (TTS) technologies. The deaf people get the voice input from the device and it displays text on the screen. The device then speaks the printed text for people with speech disabilities. This small, inexpensive device works to increase users' ability to communicate and interact independently. The device aims to provide multi-functional use, ease of use, low cost, portability, and convenience of carrying and operation in any place. In turn, it poses some challenges, like: performance heavily reliant on the quality of the components and the surroundings, dependence on the Internet to have access to Google APIs that might hinder its use in certain locations, and need for a marginal period for adaptation for new users [3].

3. Tawasul device

The Tawasul device is a smart device that enables interaction with deaf and mute people in their social and physical environment, where it is easy to convert sign language into sounds and sounds into signs that the deaf person can understand, allowing interaction and building a bridge across the communication barrier between the deaf and mute and the healthy community. In addition, it is able to detect dangerous sounds, including fire alarms or car horns, through its sign feature, thus being a communication tool and a safety device. It is small and lightweight so that a person can carry it with him for use in different environments.

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When considering the future development of this device, there has been an ongoing discussion regarding cost barriers for users, disruptions arising from inaccurate recognition of complex sign language, and reliance on battery, which makes it unusable when there is no power source. However, the communication device is invaluable in improving accessibility and independence for its users [4].

4. Smart glasses that translate sounds into sign language

Fig. 2 shows that these smart glasses use real-time speech-to-sign language translation and are among the latest technologies that can help deaf and mute people with communication challenges. With built-in microphones to capture spoken words, the glasses integrate artificial intelligence for speech recognition and sign language translation, which is displayed on a small screen directly in front of the users. This will allow people with hearing loss to engage in conversations independently, and is cited as one of the great supports for inclusive communication. They are portable, lightweight, provide real-time translation, allow for high accuracy in speech recognition, and offer flexibility towards different dialects and accents. However, they come at a high cost, with short battery life and low applicability with audio noise in global environments. Regardless of these challenges, smart glasses are a modern and effective way to give people with hearing loss much better accessibility and independence in different social and work environments [5].



Figure 2: Smart Glasses Project [5]

5. Project “Talk to Me”

“Talk to Me” is an innovative project that aims to enhance communication between deaf and mute individuals and the rest of the community through a smart application powered by artificial intelligence. The project translates sign language into written text or spoken words, facilitating the understanding of the needs of people with hearing disabilities. In addition, the application offers the feature of converting written text into sign language using a 3D character (Three-Dimensional) that realistically illustrates the signs. This project aims to enhance social interaction and integrate deaf and mute individuals into society. The project

Smart Speaking Signs Device

is characterized by its comprehensive translation capabilities, attractive interface, and AI-driven accuracy, making it an effective tool for social integration and learning. However, challenges include high development costs due to AI and 3D processing, reliance on high-performance hardware, and potential limitations in translation if the AI lacks the ability to recognize a diverse dataset. Despite these challenges, “Talk to Me” offers a dynamic and easy-to-use solution that bridges communication gaps and promotes inclusivity [6] .

2.3 Comparison With Related Work

Table 1 shows the table compares different projects that help communicate with the deaf and mute, where the technologies used vary between artificial intelligence, cameras, sensors, microphones, etc., which affects performance and ease of operation. From a cost perspective, smart glasses are the most expensive, and mobile applications are the least expensive. By developing a smart glove connected to an application, the SSSD project achieves these goals in an accurate, low-cost, and easy-to-use way. This thus provides a practical and flexible solution compared to other projects because it allows for future expansion and the addition of new languages and signs.

Criterion	Project 1: Sign Language Conversion Using Camera and AI	Project 2: IoT-Based Device for Deaf, Dumb, and Blind People	Project 3: <i>Tawasul</i> Communication Device	Project 4: Smart Glasses for Translating Sound into Sign Language	Project 5: <i>Talk to Me</i> Application	SSSD
Technology Used	Camera + Artificial Intelligence	Raspberry Pi + Google APIs	Sensors + Alert Signals	Microphones + Artificial Intelligence	AI + 3D Virtual Character	Smart Glove with Sensors + Mobile App (Bluetooth/Wi- Fi)
Primary Function	Converts sign language into audible speech	Converts text to speech and vice versa + signal display	Translates signs into sound + alerts	Real-time conversion of speech into sign language	Translates signs into text and speech	Converts gestures into audible speech + text on a computer screen
Required Devices	Camera + Computer	Raspberry Pi + Microphone + Display	Small portable device + Speakers	Smart Glasses with built-in microphone	Smartphone	Smart glove with sensors + Computer or Mobile App
Cost	High	Medium	Medium to High	Very High	Relatively Low	Low to Medium (affordable design and components)
Operating Environment	Requires well-lit and quiet environments	Requires internet connection	Works in all environments + alerts	Poor performance in noisy environments	Works in all environments	Works in all environments without limitations
Outputs	Audible speech	Audible speech and text display	Audible sound + alert signals	Immediate visual signs	Audible speech + written text	Audible speech + written text on a computer or app screen

Table 1: Compare SSSD and Related Projects

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When compared to similar existing solutions, the SSSD project has many advantages: it is easy to get, affordable, and is rather a low-to-medium-cost antithesis to the exorbitantly priced gadgets like smart glasses. It is simple and highly customizable, so it can be worked with by illiterates or firstly by those people who have technical skills. Although some other solutions do this, our solution works consistently in a variety of settings without requiring either ideal lighting or the presence of the internet to be active. Dual outputs join the written text and spoken speech, providing users a flexible way to communicate that is practical too. Another advantage is endless possibilities of adding new languages and gestures, making it user-oriented and more customizable. Our smart glove is so lightweight that it is truly suitable for daily use and is more convenient than fixed or bulky systems. SSSD project thus provides a practical, affordable, scalable, and real-time solution. This is a very impactful solution that poses serious competition to existing solutions, putting together smart sensors, gesture-to-speech conversion, and text display.

CHAPTER 3: METHODS AND MATERIALS

The smart glove must integrate highly accurate sensors to ensure precise detection of hand gestures, minimizing errors caused by unintended movements. The glove's design should prioritize lightweight and ergonomic features, allowing users to wear it comfortably for extended periods without strain. The battery must support continuous operation for 8-12 hours, ensuring uninterrupted usage throughout the day. The Bluetooth module should provide a reliable connection with a range of at least 10 meters, facilitating seamless communication between the glove and connected devices. Additionally, the accompanying app should be intuitive and user-friendly, catering to users of varying technical proficiency levels to ensure accessibility and ease of use.

In the following chapter we'll go through all of the hardware components and software tools that will be used in our system. In addition, we will discuss the flowchart and block diagram that our system uses to perform its functions, and we will program the system using that flowchart.

3.1 System Design and Components

Section 3.1 demonstrates the main hardware and software components of our system, how we selected them, and includes a block diagram that illustrates the overall system structure.

3.1.1 Block Diagram

The functionality of the system is shown in the Fig. 3. The microcontroller (ESP32) acts on input data, and it itself is a "controller". The input system, which includes the Flex Sensors and IMU Sensor, recognizes the hand gestures and the movement. The ESP32 processes the signals to convert them to a compressed digital code. The data is wirelessly minted using the built-in Bluetooth technology in the ESP32 and sent to the mobile application which is developed using Flutter. The mobile application receives the code, and then connects with the Firebase for an appropriate gesture. After establishing the connection, if the system found the gesture on the post, the appropriate text appears on the display screen and provides the audio output. The system is fully integrated; the glove, the ESP32, the mobile application, and the Firebase all work effortlessly together to provide easy communication and instant recognition. It is especially helpful for the deaf and mute communities in public locations, and places like hospitals.

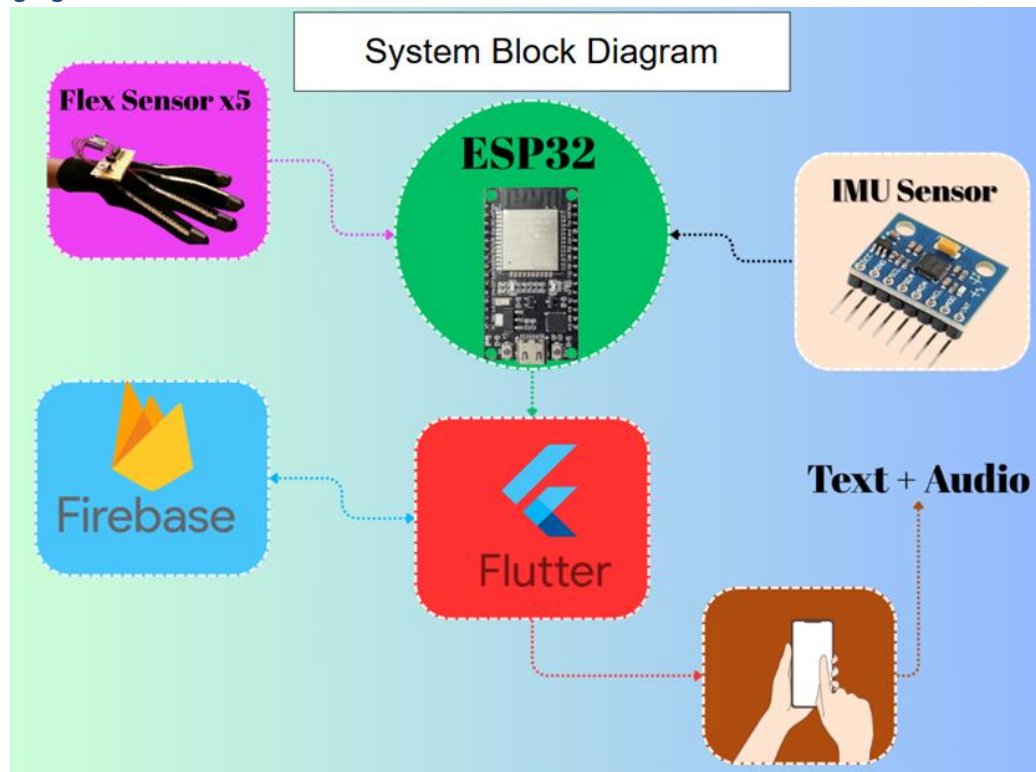


Figure 3: Block Diagram

3.1.2 Hardware Components

- A. Microcontroller (ESP32).
- B. Glove with Flex Sensors.
- C. Glove with IMU Sensor.
- D. DC to DC Step-Up Booster Module.
- E. TP4056 Li-Ion Battery Charging Module.
- F. Li-Ion Battery.
- G. PCB Board.
- H. USB Data Cable.
- I. Fixed Resistors.
- J. Jumper Wires.
- K. Laptop.
- L. Samsung Smartphone.

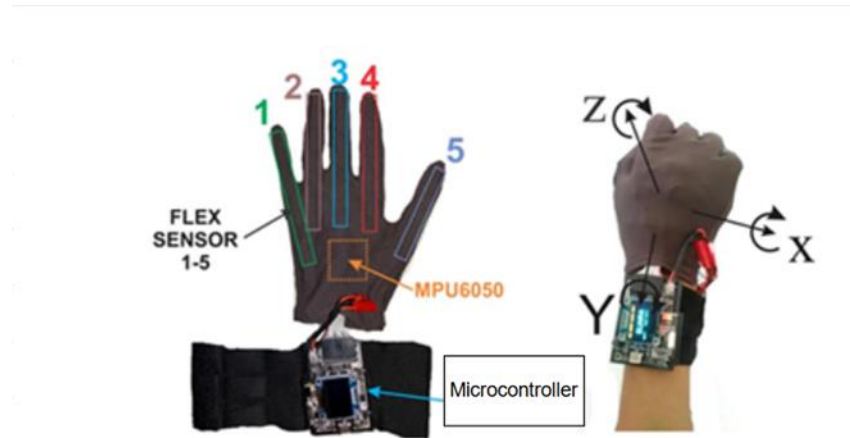


Figure 4: Attachment sensors on smart glove [7].

A. Microcontroller (ESP32):

Fig. 5 demonstrates that ESP32-WROOM-32 is a high-performance microcontroller used for precise control in smart systems. It has a fast dual-core processor and onboard Bluetooth and Wi-Fi and is suitable for IoT and wearable technology applications. The board has a small size, low power consumption, and has many digital and analog input/output pins. It is easy to develop on the Arduino IDE and has a low voltage level of 3.3V, making it safe and efficient to connect with other electronics.

The ESP32 Dev Kit V1 board, whose focal point was the ESP32-WROOM-32 module, was used in our project, this board receives input from the flex sensors of fingers and the MPU6050 sensor. It converts the above information into understanding the hand gesture made by the user. It forwards the result to a mobile app through Bluetooth. The application shows the result in text and voice-announces the result. The ESP32 is attached to a PCB board and is driven by a rechargeable Li-Ion battery, thus rendering the glove wearable, portable, and handy to use in everyday tasks [8].



Figure 5: ESP32 [9].

B. Glove with Flex Sensors:

Fig. 6 shows the flex sensor that can be used to sense bend or flex of things. It is suitable with a voltage ranging from 3.3 to 5 volts. That is ideal for the smart gloves that are now common in wearables since it can detect finger movement and bending that can easily be interpreted into useful function like sign language translation [10].

The smart glove is designed as an integrated device that includes five Flex sensors attached to each finger, working in conjunction with IMU sensors. The Flex sensors are used to detect finger movements and accurately measure the flexion of each finger, based on changes in electrical resistance as the finger flexes. These sensors are ideal for tracking motion because their resistance changes based on the angle of flexion. Meanwhile, the IMU sensors measure motion across three axes (X, Y, Z), providing accurate data on acceleration, velocity, and angle. The sensors work together in parallel to provide a complete picture of hand gestures and movements. The Flex sensors capture data to determine the degrees of flexion for each finger, while the IMU sensors complete the measurements of the overall hand motion. All data is sent to the controller for real-time analysis to determine the gestures made by the user. For example, if the user flexes their fingers and moves their hand in a certain pattern, the Flex sensors capture the degrees of flexion for each finger, while the IMU sensors track the overall hand motion. This data is then analyzed and converted into written text or spoken words via the app. This integration enhances the accuracy of gesture recognition and provides immediate feedback.



Figure 6: Flex Sensors [11].

C. Glove with IMU Sensors(MPU6050):

Fig. 7 shows the MPU6050 sensor, which consists of two main components: an accelerometer and a gyroscope. It has an operating voltage of 3.3 to 5 volts and is used extensively in systems requiring precise motion tracking [12].

The smart glove is introduced as a distinct technology encompassing sensors that focus on tracking gestures made by the hand and the finger movements alongside it. The smart glove is equipped with five IMU sensors, such as (MPU6050), the devices that can perform movement measurement in (X, Y and Z) degrees using accelerometers, which measure acceleration along these axes and determine the direction of movement, such as raising or lowering the hand. Additionally, the gyroscope measures angular velocity around the three axes, allowing it to detect rotation or twisting movements, such as wrist rotation. These sensors are designed to track gestures and the finger movements, which are then delivered as input to a microcontroller in real-time. For instance, when a certain hand gesture is made by a user, the hand, finger sensors that are affixed within the glove grasp the motion and process the information. The mobile application that connected together with the glove via the communication Module wirelessly translates the information to text and speech.

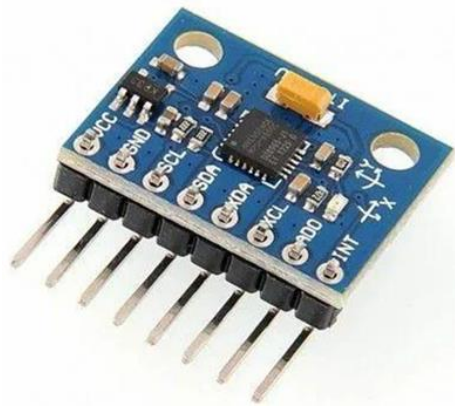


Figure 7: MPU6050 Sensor [13].

D. DC to DC Step-Up Booster Module:

Fig. 8 shows our DC to DC Step-Up Booster Module that was soldered on the PCB and other crucial components. The module voltage increases from a low-power source, e.g., 3.7V lithium battery, to a stable higher output of, for example, 5V. It efficiently boosts voltage with minimal energy loss and is thus a good choice for powering modules like the ESP32 and other sensors, which have higher operating voltage requirements. In our project, the booster module gives a guaranteed and stable supply of power to all the devices connected to it, which is very necessary for the stable operation of the ESP32 microcontroller



Figure 8: DC to DC Step-Up Booster Module [14].

E. TP4056 Li-Ion Battery Charging Module:

Fig. 9 shows the TP4056 charging module, a compact and efficient method of charging 3.7V rechargeable lithium-ion batteries through the constant-current/constant-voltage (CC/CV) charging mode. It contains a Mini-USB port that can be directly plugged into the computer USB port or energized from an external power source through the IN+/IN- pads at a voltage of 4.5–5.5V. The module is supplied with the integrated TP4056 chip and charges a maximum of 1A. It also contains LED indicators for the charging status: red when charging and green when charged. The module is miniaturized ($17 \times 22 \times 5$ mm) and lightweight (5 grams) and is therefore appropriate for installation in portable and wearable products.

We utilized the TP4056 module to charge a lithium-ion battery, which we utilized as the main power source for the whole system. Given that this module has automatic charging control and visual indication, charging the battery was made safe and effective. As such, we were able to do away with large external chargers, thus making the glove lighter, more portable, and easy to use on a daily basis [15].

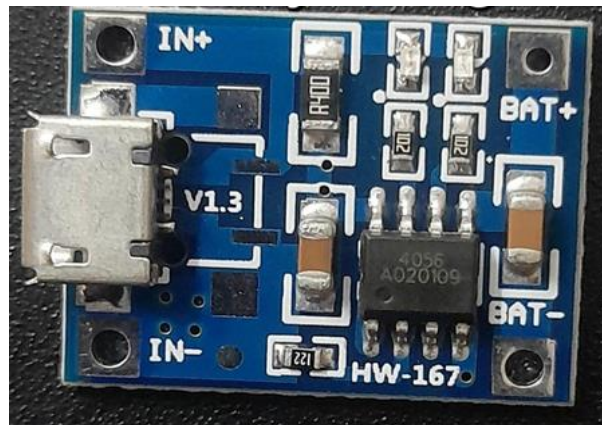


Figure 9: Li-Ion Battery Charging Module [16].

F. Li-Ion Battery:

Fig. 10 shows a lithium-ion battery, which ions move from the negative electrode to the positive during discharge and vice versa when filling. Between the two electrodes there is a separator (partition that separates them), and everything together is immersed in an electrolyte that facilitates the movement of ions. Their rated voltage is 3.7V (3.6V). And it is also a long life battery.

We chose to utilize an 18650 lithium-ion battery as our power supply for our project. We mounted the battery onto the glove and connected it to a TP4056 lithium-ion charging module for the battery. TP4056 regulates the charging and prevents overcharging, and therefore it is a safe and effective option for our

Smart Speaking Signs Device

wearable. The battery voltage output is then input into a DC to DC Step-Up Booster, which boosts the voltage to 5V — the level required to power the ESP32 and other sensors. Such a system provides a safe and convenient power supply, yet maintains the glove light and easy to use without being plugged into an external source [17].



Figure 10: Li-Ion Battery [17].

G. PCB Board:

Fig. 11 shows a printed circuit board (PCB) is a flat board used to hold and connect electronic components electrically by soldering them into conductive paths. The base layer of a PCB is typically made from a combination of strong, nonconductive materials that provides insulation, water resistance, and temperature stability. Etched onto the base layer of a PCB are pathways that carry signals from point to point. Called “traces,” these thin bodies are usually made of copper, a highly conductive material that allows electrons to move with little resistance between components. It is extremely easy to work with and you can easily solder on this board as it has soldering spots and some gap between them to avoid short circuiting

We used a PCB board in our project to join and hold the key components such as the ESP32 module and other components in place. The resistors and wires were soldered on the board directly to create a complete and rigid circuit. This gave the glove a clean appearance with an easy installation, and also ensured that the wires were not likely to disconnect or the connections break during normal usage. Through the use of the PCB, the glove became rigid and organized, and allowed us to create a wearable and functional prototype [18].

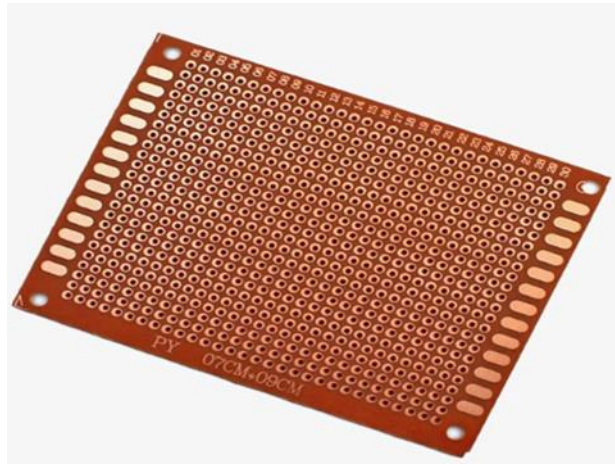


Figure 11: PCB Board [19].

H. USB Data Cable:

Figure 12 depicts the Y OSI Series USB data cable used as part of our project. This cable had two main functions: it was used to upload the program code from the laptop to the ESP32 microcontroller during the development and testing phases, and it was used to establish the serial connection between the ESP32 and laptop for monitoring the sensor readings and debug data using the Serial Monitor in the Arduino IDE.

The Y OSI Series cable also had power delivery in which we used power from the laptop to temporarily power the glove when testing at the beginning stages. There was also the possibility to plug it into some modern smartphones when needed, due to it being compatible with USB Type-C. Using the Y OSI Series cable was easy, flexible, and high quality, which contributed to the development process and demonstration of the system.



Figure 12: USB Data Cable [20].

I. Fixed Resistors:

Fig. 13 shows a 10k Ω fixed resistor used in the hardware setup of the smart glove system. These resistors were connected in voltage divider circuits alongside the Flex Sensors. Their main role is to convert the varying resistance of the sensors—caused by finger movements—into voltage signals that can be accurately read by the ESP32 microcontroller.

This configuration enables precise detection of finger bending, which is critical for interpreting hand gestures. The 10k Ω value was selected for its optimal balance between sensitivity and stability, ensuring consistent readings that improve the overall reliability and responsiveness of the system.



Figure 13: Fixed Resistors [21].

J. E-wires:

E-wires is a set of Wires are thin, flexible strands of metal that are used to carry electrical current from one point to another. They are typically made of copper or aluminum, due to their high conductivity and low resistance to electrical current. Wires commonly used in a wide range of applications, including electrical power distribution, electronic circuit connections, and telecommunications. They are usually insulated with a protective coating of plastic or rubber, to prevent them from coming into contact with each other or with other conductive materials. Different types of wires are designed for specific purposes, such as high-voltage power transmission, low-noise signal transmission, or high-frequency data transmission. Fig. 14 displays the jumper male cables [22].



Figure 14: E-wires [23].

K. Laptop:

The laptop was set as the focal point of the project development setup. It was used to program the ESP32 microcontroller with the Arduino IDE, upload codes, test sensor readings, and debug processes. It was also used as the focal platform to develop and run the Flutter mobile app. This offered complete control over the hardware and software development processes in one setup.

L. Samsung Smartphone:

The Samsung smartphone was used to simulate and test the Flutter application designed to interface with the glove system. It received gesture codes from the ESP32 via Bluetooth, matched them with the stored data in Firebase, and displayed the corresponding translated text. Furthermore, it provided audio output for enhanced communication, making it the main user interface during both testing and demonstrations.

M. USB-type C:

The USB Type-C cable is a high-speed data and power transfer cable that is modern and widely used with laptops and smartphones. In our project, it was employed to create a direct link between the Samsung smartphone and the laptop. This link permitted us to install, execute, and debug the Flutter application in real time from the laptop development environment.

3.1.3 Software Components

- A. Mobile Application.
 - Flutter.
- B. Firebase.
- C. Arduino IDE.

A. Mobile Application:

The mobile application serves as the direct communication element between the user of the system and the system itself. The app is developed using Flutter (a cross platform framework) which enables direct communication with Firebase, and also has quick access to Bluetooth connections.

In the SSSD system the smart glove communicates with the mobile application via Bluetooth. The ESP32 will send a unique code for the corresponding gesture via Bluetooth. The mobile application receives the code for the gesture and will check the Firebase Realtime Database to match the code to the particular sentence or word and its associated audio.

Once matched the application will clearly define the translated gesture as a written text sentence, and will play the audio that corresponds in an attempt to assist communication between the participant and facilitator. The user interface was developed using Flutter, making it easy, intuitive, for interaction with other functionality (changing language, sound on/off, notifying user if the application has not recognized their gesture).

This connects the application with the smart glove (the hardware), and the Firebase database (a cloud backend), and allows the application to effectively and accurately translate the action of the gesture to the written text and audio in.

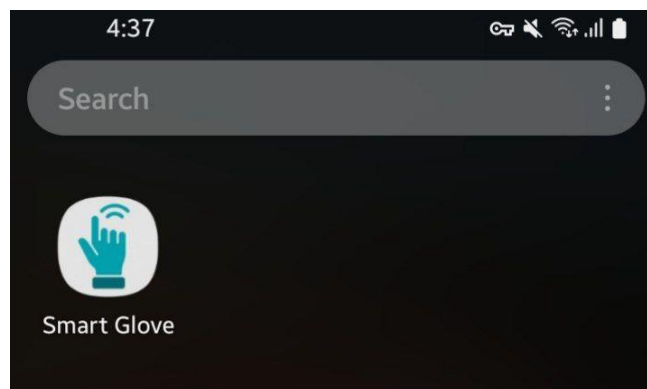


Figure 15 : Mobile app icon

- **Flutter:**

Flutter is an open-source framework from Google that is written in Dart language for building natively compiled applications for mobile (Android and iOS), web, and desktop from a single codebase. In this case of the SSSD system, the UI of the mobile application is developed based on the Flutter framework in the smart glove system, ensuring a smooth and responsive operation in the SSSD system.

One of the things I love most about Flutter is its hot-reload capabilities – you make a change and you can see the result in a running app without having to restart it. This speeds up the workflow enormously, when new gesture based functionality is going to be added or changed. High level of compatibility with the framework and good integration with Firebase services and Bluetooth packages make it fast and responsive.

It's not all sunshine and rainbows however: Flutter apps tend to be relatively heavy compared to their native counterparts, which is not good for budget devices.

But with its rapid iteration process, accessibility to new developers and ability to adapt with the latest designs, Flutter remains a good candidate for building apps which in turn can help facilitate communications between people who have hearing and voice impairments and the world at large, thus furthering understanding and accessibility [24].



Figure 16: Flutter [25].

B. Firebase:

In the SSSD system, the database (Firebase) serves as a mediator in the translation of gestures from the smart glove to the relevant output code for the control signal in the instance of the hand gestures code. The control signal of a hand gesture in code form is picked up and sent through Bluetooth through the ESP32

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to the mobile application. The application would then query the Firebase Realtime Database to get the code, and output the relevant phrase or expression.

Firebase is mainly the data access layer of the SSSD system. It does provide access to the real-time lookup of gesture codes associated with text outputs. Therefore, the glove and application respond in real-time and are synchronized.

This project does not use Firebase for hosting of audio, as would be the case for systems that actually have saved files of audio that are typically Pre-recorded. The mobile application utilizes a Flutter package called Text-to-Speech (TTS). This will take the text code retrieved and output the spoken version directly on the device. It also allows for a smaller size storage option and is also less complicated of design and internal architecture than systems that use Firebase for their audio tools and then have to follow the traditional way of collecting and rendering spoken sounds.



Figure 17: Firebase [26].

C. Arduino IDE:

The Arduino IDE was the main platform to program the ESP32 microcontroller that is the main processing unit of the smart glove in the SSSD system. The required code was written in the Arduino IDE and uploaded to the ESP32 to read information from the sensors (the flex sensors, and the MPU6050 motion sensor).

The Arduino code handled an important task in establishing Bluetooth communication by making use of the ESP32's built-in Bluetooth module which allows the microcontroller to wirelessly transmit gesture codes to the mobile

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application. The Arduino IDE is a user friendly and a very useful way to write, upload and verify code which was useful for gesture detection, and wireless transmission of information.

While initializing the Bluetooth module, the ESP32 will constantly read information from the sensors, processes the information and send the correct gesture code to the mobile app via Bluetooth. In such a way the glove and mobile application can communicate with each other real time.

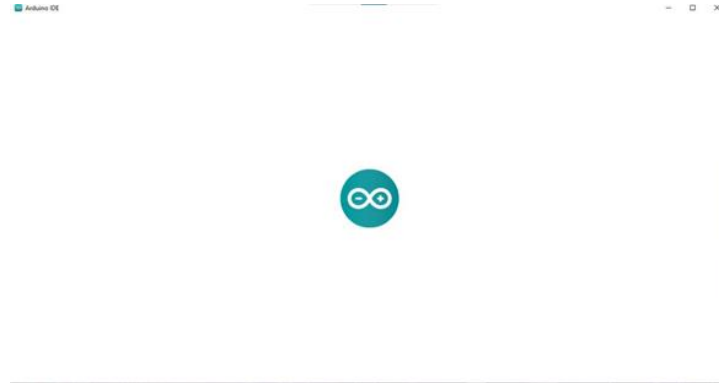


Figure 18: Arduino IDE.

3.2 Design Specifications, Standards and Constraints

3.2.1 Specifications

These are the most significant hardware specifications for the design process. The operating temperature is important because it determines whether the sensors will work under different environmental conditions. Another key factor to consider is the size of the components to ensure they fit well within the glove design and are portable.

Component	Operating Temperature (°C)	Size (mm)
Gloves	NA	Adjustable
IMU Sensors (MPU6050)	-40 to +85	20 × 15 × 2
Controller (ESP32)	-40 to +125	51 × 25 × 10 (approx.)
Flex Sensors ×5	-20 to +85	50 × 6
Smart Phone	0 to +50	Varies
Li-Ion Battery (18650)	-20 to +60	65 × 18 (typical)
DC to DC Booster	-20 to +85	42 × 31 × 14 (typical)
TP4056 Charging Module	-10 to +85	22 × 17 × 5
Fixed Resistors	-55 to +155 (typical)	Standard sizes

Table 2: Components Specifications

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As we can see in table 2, the system's components are suitable for varying environmental conditions, with a minimum operating temperature of -40°C and a maximum of $+85^{\circ}\text{C}$ for most sensors. Moreover, the sizes of the components are compact, making them suitable for integration into the glove design.

3.2.2 Standards

1. Bluetooth Communication: This standard ensures secure and power-saving data transfer between the ESP32 microcontroller and the smartphone application. We need to ensure stable and power-saving data exchange between the Mobile app and the ESP32 microcontroller.

2. Software and Application Development: This standard focuses on software quality, including efficiency, usability, and reliability. It ensures that the smartphone application provides a smooth and effective user experience.

3. Safety of Wearable Devices: This standard addresses wearable device safety, that the glove will be safe if used for a long time and will not be dangerous to the user under normal use.

3.2.3 Constraints

- The first limitation is ensuring sensor accuracy, as the bend and motion sensors must detect gestures with high precision, and unintended movements could cause errors in the system.
- Second, the battery life is a constraint, as the glove needs to operate continuously for 8-12 hours, requiring careful power optimization.
- Third, ease of use is crucial since the glove and its accompanying app must be intuitive and accessible, especially for users who are not tech-savvy.
- Fourth, the size and weight of the glove are limitations because it must be lightweight and ergonomic to ensure user comfort during extended use.
- Fifth, the Bluetooth connection range is a constraint, as the glove must maintain a reliable connection within at least 10 meters to avoid disconnection issues.

3.3 Design Alternatives

In the Alternatives section, we explain the modifications made to certain components within our project, as well as an introduction to the newly integrated component.

We initially used the Arduino Uno as our main controller, but due to its large size and lack of wireless connectivity capabilities, we replaced it with the ESP32. The ESP32 boasts better processing capabilities, Bluetooth and Wi-Fi connectivity integrated into it, and is compact compared to the Arduino Uno and sits well in the glove. Thanks to its onboard wireless functions, the ESP32 also eliminated the need for external special communication modules such as the external HC-05 Bluetooth module.

As for gesture detection, we selected flex sensors over other alternatives such as capacitive or camera-based sensors due to their flexibility, ease of integration into wearable fabrics, and accurate . Flex sensors are lightweight, low-cost, and not affected by lighting or environmental conditions, making them a practical and reliable choice for continuous finger motion tracking in a wearable system like our smart glove.

These innovations made the SSSD system more efficient, portable, and user-friendly overall and more wearable-friendly.

3.4 System Analysis and Optimization

3.4.1 System Requirements

Functional Requirements:

1. **Convert gestures to text and speech:** The system translates gestures captured by the glove into written text displayed on the app and speech in less than one second.
2. **Wireless communication between the glove and the mobile phone app:** Use Bluetooth HC-05 technology to transmit data between the smart glove and the mobile phone app within a range of 10 meters, and ensure a stable and fast connection for smooth data transmission.
3. **Database management and gesture verification:** The system must query the gesture database to find a matching entry for the detected gesture: Support flexible storage to allow easy updates and addition of new gestures in the future.

Non-Functional Requirements:

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1. **Performance Accuracy:** Sensor readings are crucial to ensure system reliability, and gestures should be translated into text and sound in a very short time.
2. **Portability and Design:** The glove should be lightweight and comfortable for long-term use.
3. **Cost:** The device should be produced at a low cost to make it affordable for a wide range of users.
4. **Reliability:** Bluetooth connection must be stable, and the system should work continuously without failures.
5. **Ease of Use:** The application should have a simple user interface that is easy for anyone to use.
6. **Scalability:** The system should support future updates to add new gestures or languages without changing the hardware.
7. **Power Consumption:** The glove should consume minimal power to ensure long battery life.
8. **Compatibility with smart devices:** A mobile app is developed to connect with the smart glove via Bluetooth. The app receives and translates data with a simple and user-friendly interface.

3.4.2 Use Case Diagram

User Use Case:

Figure 19 illustrates the interaction between the Deaf and Mute User and the components of the Smart Speaking Signs Device. The smart glove must be put on before the user can make hand gestures. After identifying these movements, the device converts them into a corresponding gesture code. The system receives the gesture code, interprets it, and generates useful output (such as text and audio). The result is then displayed by the linked mobile application, providing instantaneous visual and audio feedback. This diagram explains the main features of the system and illustrates the crucial steps from gesture input to result display.

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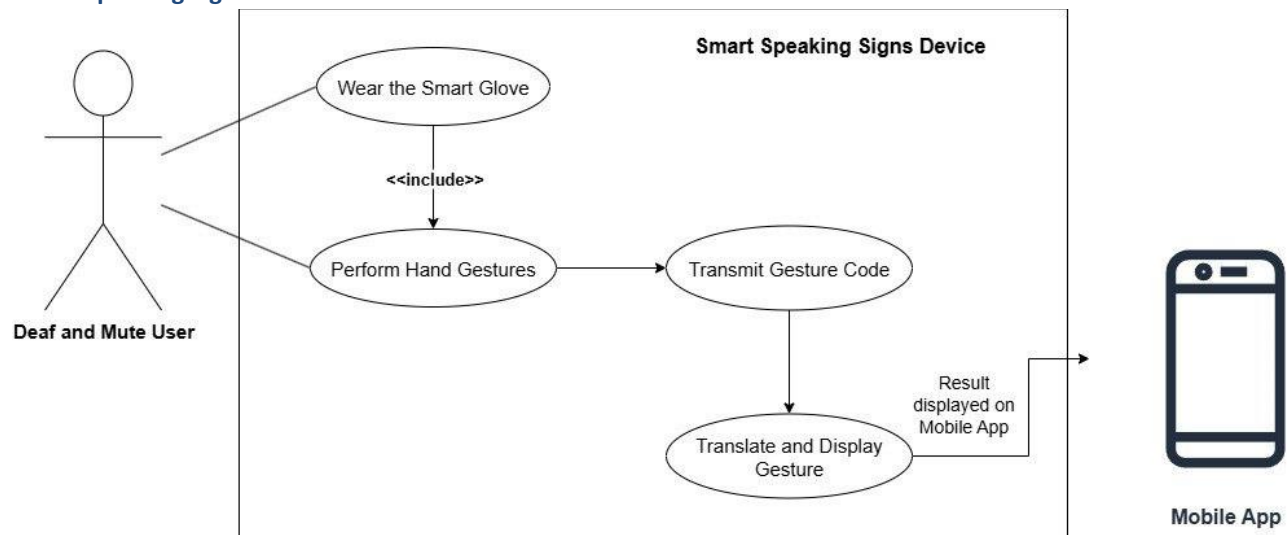


Figure 19: User Use Case.

Microcontroller Use Case:

Figure 20 illustrates the smart glove controller (ESP32), the brain of the smart voice signaling device. The controller collects real-time sensor data from the inertial measurement unit (IMU) and the flexibility sensors, processes it to generate a gesture code, and then transmits the code via Bluetooth Low Energy (BLE) to the associated mobile app.

Each function is represented as a separate use case that interacts directly with the controller. The mobile app is also shown in the diagram as an external system that receives the gesture code and uses it for additional display and interpretation. This use case model explicitly defines the built-in logic that the device handles before communicating with the app.

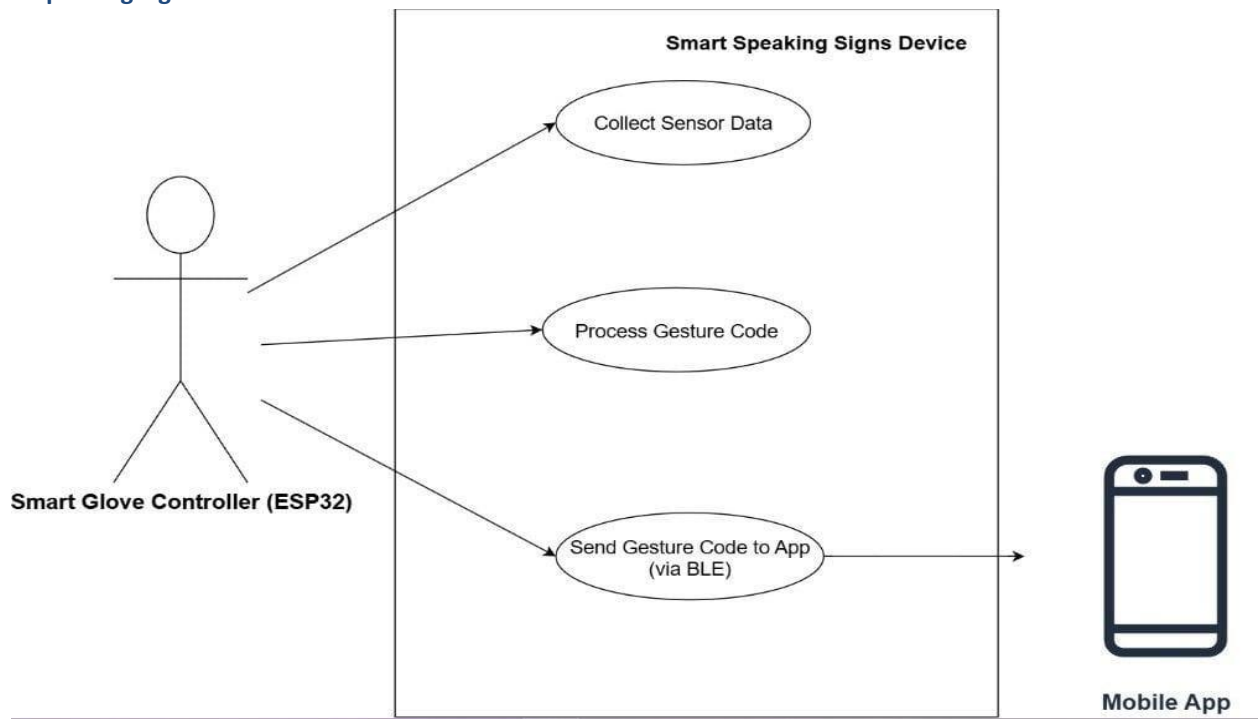


Figure 20: Microcontroller Use Case.

3.4.3 Sequence Diagram

Figure 21 shows how the deaf & mute user and the components that make up the Smart Speaking Signs Device interact. The sequence starts when the deaf and mute user wears the smart glove to make a gesture. This movement is detected by the glove's sensors, which then transmit the information to the ESP32 controller. After processing the data, this controller creates a code that represents the gesture that was performed. The mobile application then receives this code via Bluetooth Low Energy (BLE)

The application matches the code with its previously stored meaning by connecting with the Firebase Realtime Database after receiving the code. The application simplifies communication by displaying the meaning of the gesture as readable and spoken text if it is recognized. The user is informed by the system that there is no match if the gesture is not recognized. This diagram clearly illustrates the sequence of operations, starting from the hand gesture to the outcome shown on the mobile phone, demonstrating how this system contributes to supporting communication for people with hearing and speech disabilities.

Smart Speaking Signs Device

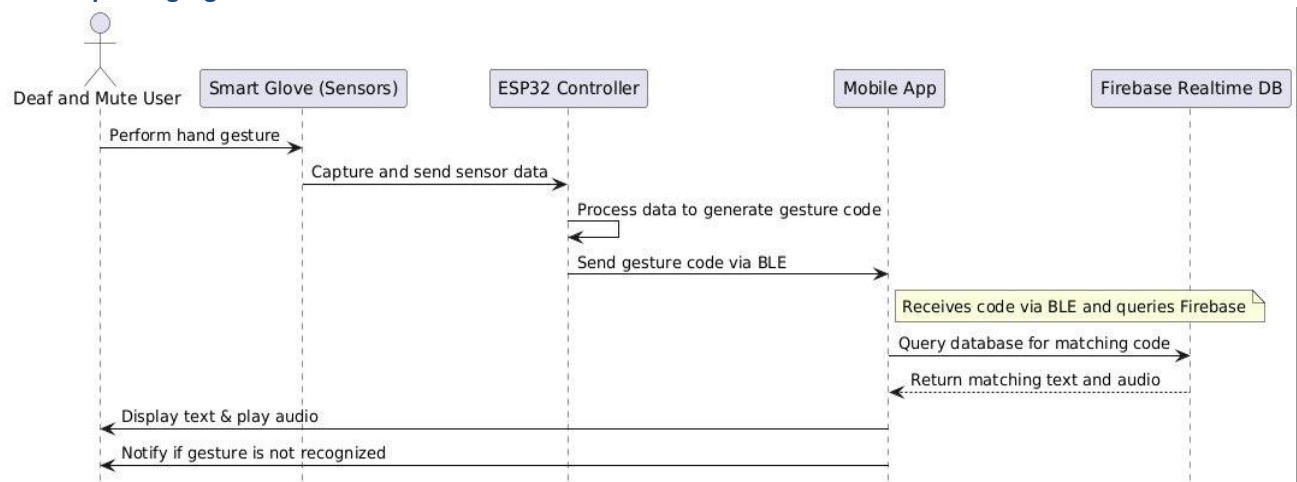


Figure 21: Sequence Diagram.

3.4.4 Flowcharts

App Flowchart:

Figure 22 illustrates the logical flow of the Smart Glove mobile application. When the application first launches, Firebase is initialized and Bluetooth permissions are verified. It then searches for an ESP32 device using Bluetooth Low Energy (BLE). After connecting to the device and learning about the services that are available, the application enrolls in the notification feature to receive action codes that have been captured by the smart glove.

Once an action code is received, the application checks whether the code represents a "no match" (00000) and displays either a suitable message (no match) or queries the Firebase Realtime. If Firebase contains the code, the meaning is shown on the user interface and spoken using the Text-to-Speech library (if voice is enabled). Additionally, the meaning is stored for later viewing in the history.

In order to ensure a real-time interactive experience between the user and the smart glove via mobile phone, the diagram also shows that the application operates in a loop while waiting for new codes from the ESP32.

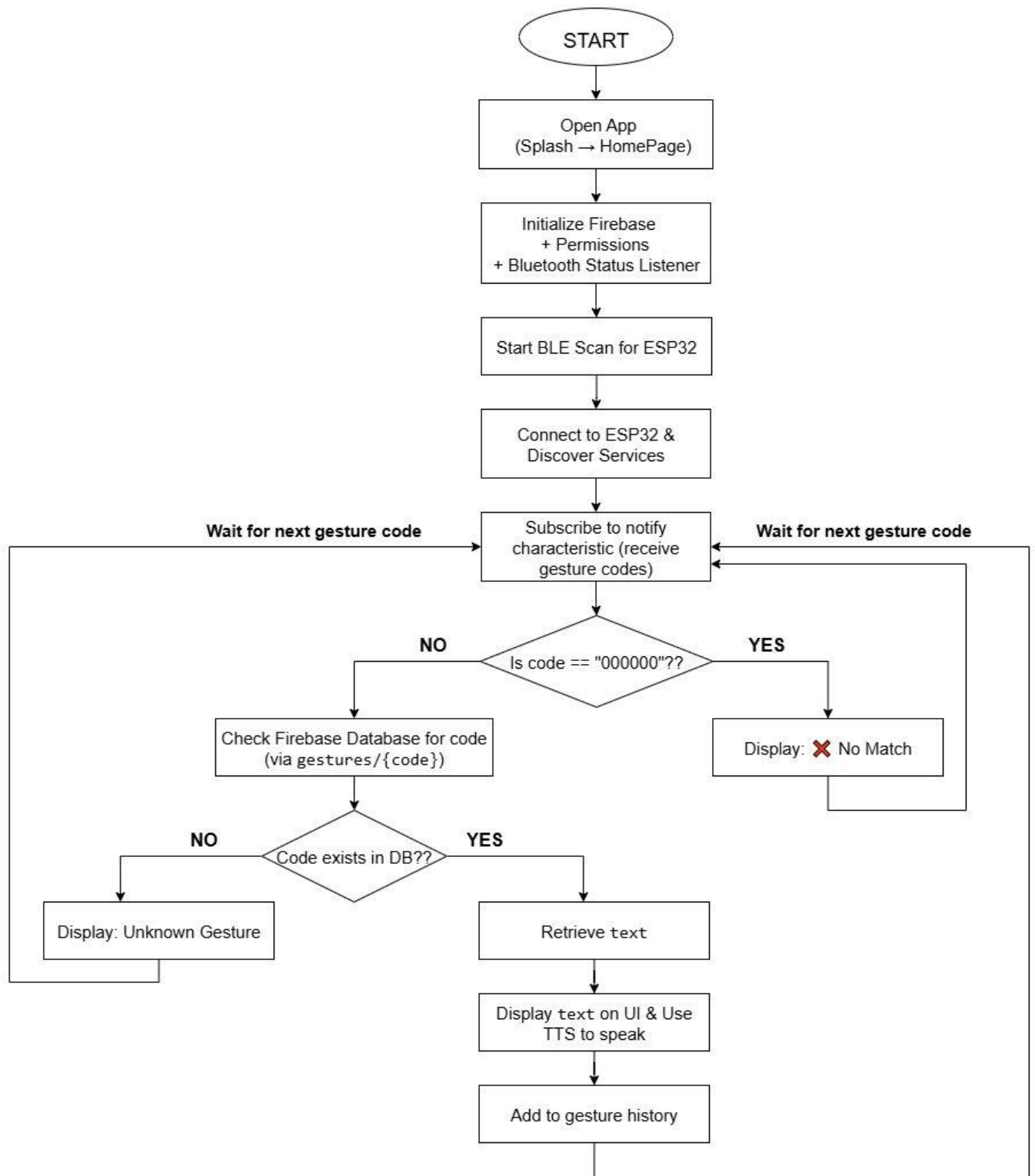


Figure 22: App Flowchart.

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Microcontroller Flowchart:

Figure 23 illustrates the operation sequence of the ESP32 controller in the Smart Glove project. The process begins by initializing the serial communication and the I2C interface to connect the sensors. Then, the controller attempts to connect to the motion sensor (MPU6050). If the connection is successful, the IMU readings (accelerometer and gyroscope) will be enabled; otherwise, they will be ignored.

Next, Bluetooth Low Energy (BLE) communication is initialized under the name "SSSD". This includes creating a BLE server, setting up its service and characteristic (with notification), and starting BLE advertising.

After that, the system goes into a timed loop where it reads data from the five flex sensors every read Interval. The system also reads gyroscope and accelerometer data if the MPU6050 sensor is available.

A predetermined list of gesture patterns is used to compare the gathered sensor data. If a match is detected, the corresponding gesture code is determined. If an Inertial Measurement Unit (IMU) is available, its data is used to increase accuracy. The match code is printed if a correct gesture is identified. By default, "00000" is used if no match is detected. Finally, the code (either a match or the default) is sent to the app to be displayed as text and audio if the device is connected via Bluetooth (BLE).

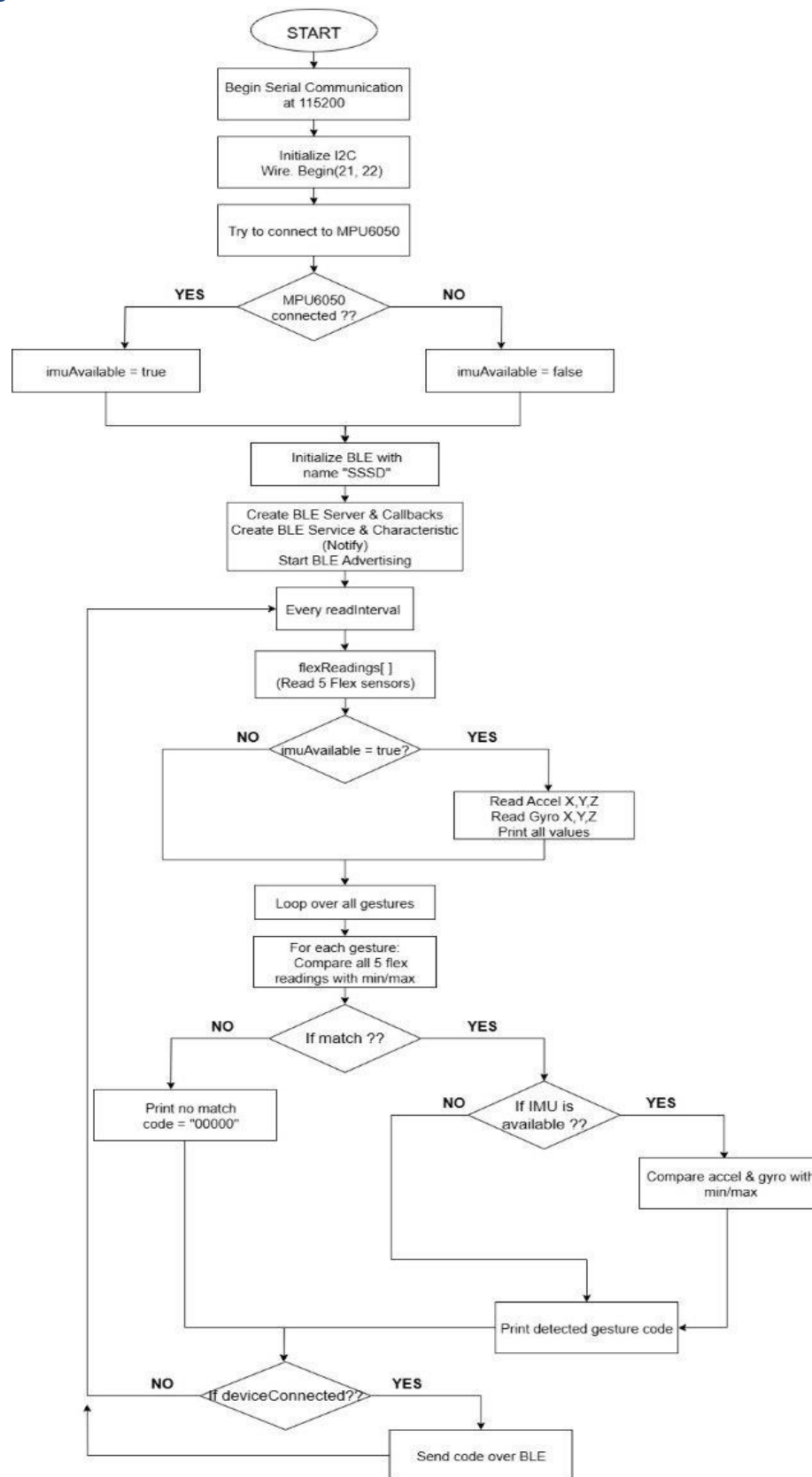


Figure 23: Microcontroller Flowchart.

3.5 Simulation and/or Experimental Test

Fig. 24 illustrates the SSSD system hardware circuit. Showcasing how the various components are connected and how they interact to enable gesture recognition and communication. The ESP32 microcontroller lies at the core of the system, collecting analog signals from five Flex Sensors attached to each finger. These sensors change resistance based on the degree of finger bending. Fixed resistors are connected in series with each Flex Sensor to form voltage dividers, and the resulting signals are read through the ESP32's ADC pins.

Additionally, the MPU6050 IMU sensor is connected via the I2C interface (SCL and SDA) to the ESP32, providing motion and orientation data. All sensor readings are processed inside the ESP32, which runs a built-in algorithm that converts the combined sensor data into a specific 5-digit gesture code.

Once a code is generated, the ESP32 uses its built-in Bluetooth module to transmit this code to the Flutter-based mobile application. The app then queries Firebase Realtime Database to find a match for the received code. If a match is found, the application displays the corresponding word or phrase both as text and audio, enabling smooth and intuitive communication for deaf and mute users.

The entire system is powered by a Li-Ion 18650 battery, which is charged using the TP4056 module. A DC to DC step-up booster raises the battery voltage to 5V to meet the ESP32's requirements. The modular design and portable power setup allow for a lightweight, wearable, and wireless operation ideal for daily use.

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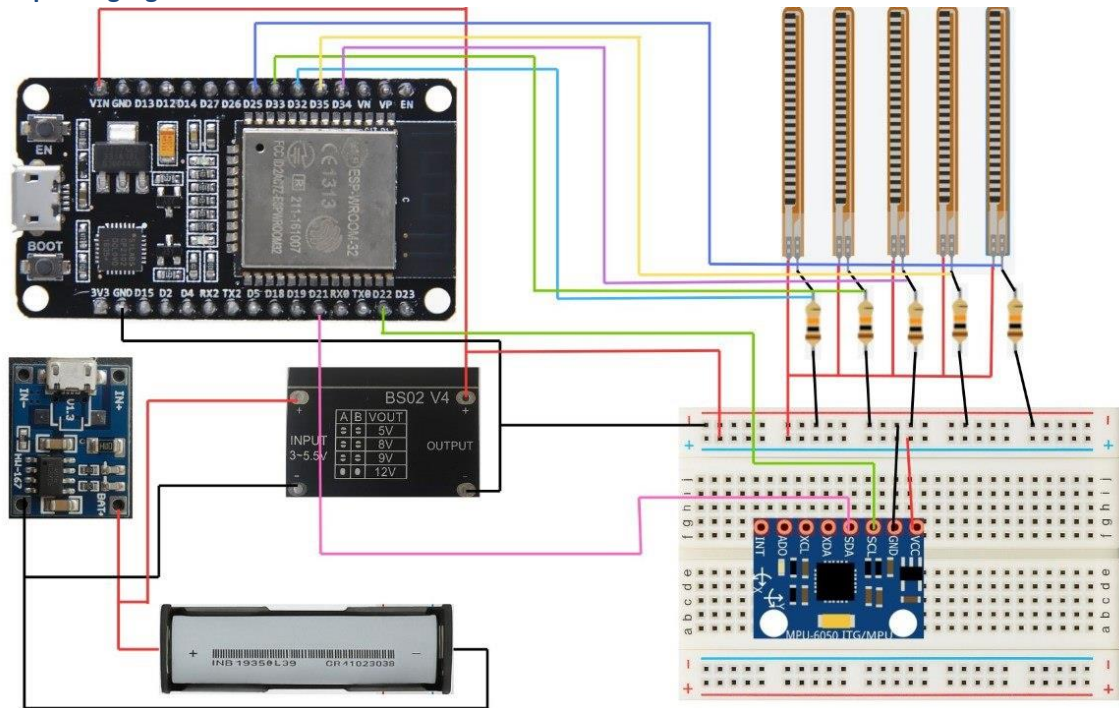


Figure 24: Component Simulation.

CHAPTER 4: RESULTS AND DISCUSSIONS

4.1 Results

In this section, we will test the project to ensure that it gives us the results we want to achieve or not.

Hardware Results:

The hardware setup with ESP32, flex sensors, and MPU6050 is shown in Figure 25. The system was assembled to detect hand gestures and send them via Bluetooth to the mobile application



Figure 25: Hardware Results-1

The flex sensors and MPU6050 motion sensor were attached to the ESP32 microcontroller and a code was written via the Arduino IDE to read and process the sensors. More specifically the flex sensors were measuring the amount of bending at each finger and the MPU6050 was measuring the capture of

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acceleration and rotation in overall hand movement. Figure 26 reveals the raw readings captured from the both flex sensors and MPU6050 during a static test of the glove.

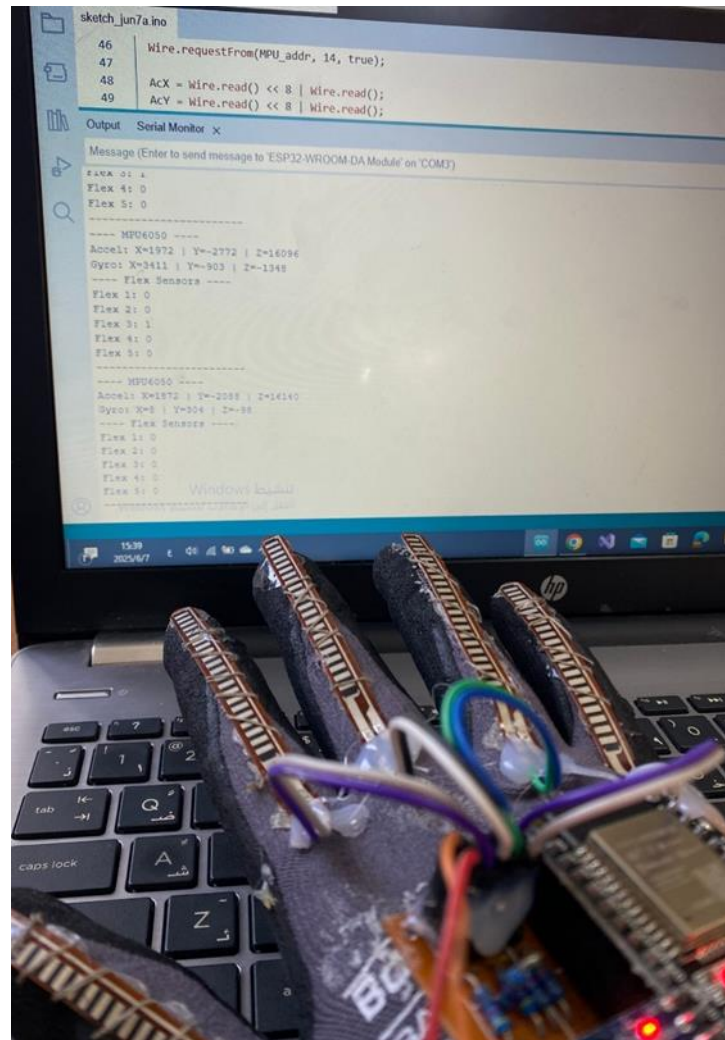


Figure 26: Hardware Results-2.

The connections of the smart glove components with the ESP32 microcontroller are shown in Figure 27. This image demonstrates how the ESP32 controller serves as the main communication bridge between the hardware and the mobile application using Bluetooth technology. Unlike earlier setups where data was only displayed locally through Arduino, the ESP32 enabled real-time wireless transmission of motion data from the glove to the mobile app. This connection allowed the system to recognize gestures and display the corresponding output by matching the received codes with stored values in the Firebase Realtime Database.

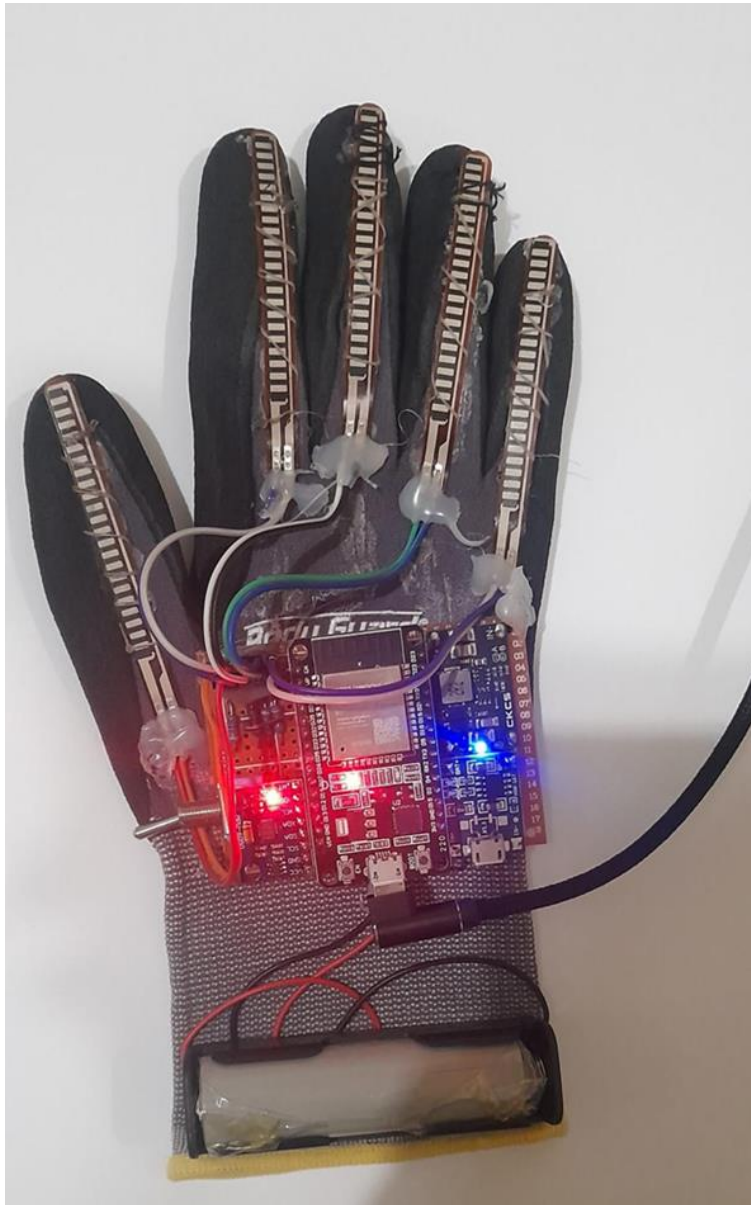


Figure 27: Hardware Results-3.

Fig. 28 provides an overview of gesture detection in which the smart glove readings, using the five flex sensors and the MPU6050 sensor, are detected, processed, and sent via Bluetooth to the mobile app. The mobile app sends a unique code for the detected gesture, and then queries the Firebase Realtime Database to find that code. Once the code is found the app will show the appropriate phrase on the screen. For example, the code "00001" is sent from the ESP32, which translated to the phrase "شكراً" can be retrieved from Firebase and shown to the user in text or in voicing using the Text-to-speech function.

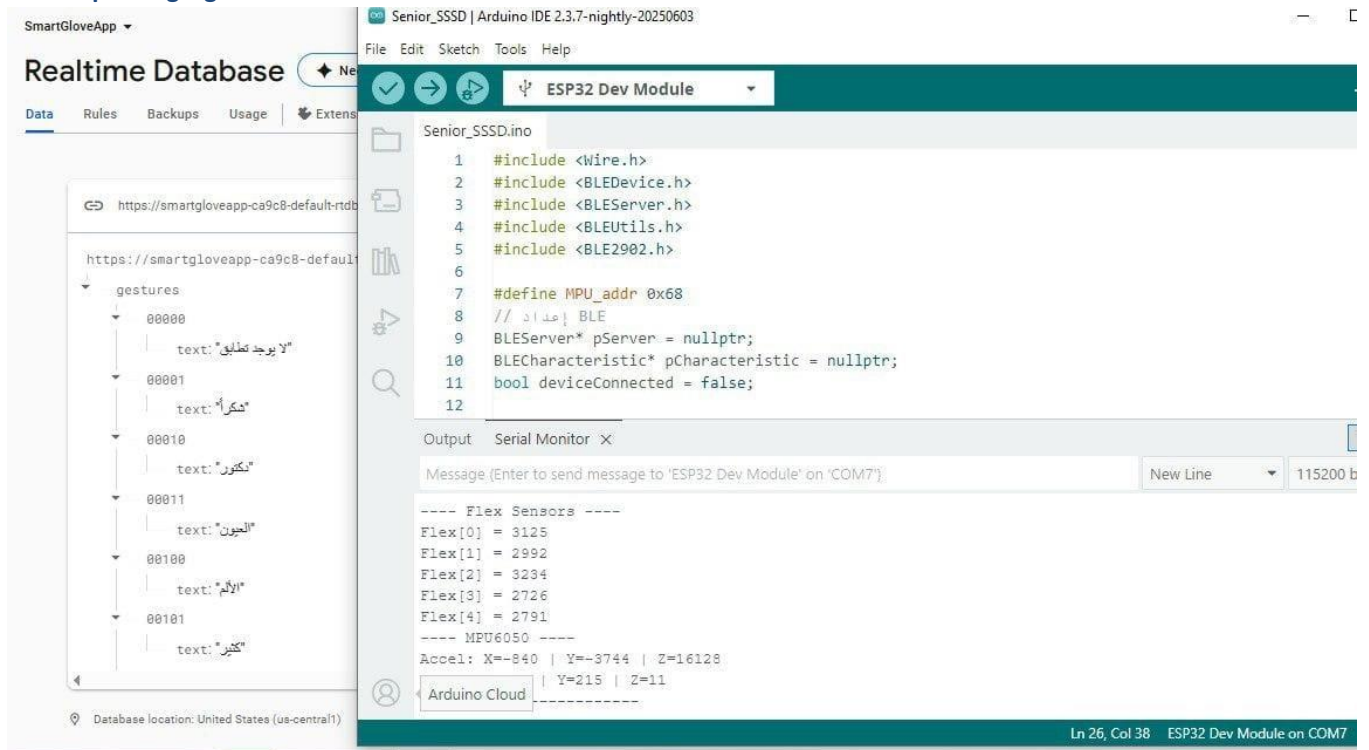


Figure 28: Arduino IDE result.

Fig. 29 shows the final design of the SSSD system, including the smart glove connected to the ESP32 board with all sensors fully integrated. The flex sensors are attached to the glove fingers, while the MPU6050 sensor is fixed to track hand orientation. The setup also includes the Bluetooth module for wireless communication with the mobile application, which receives and processes the gesture codes in real-time.

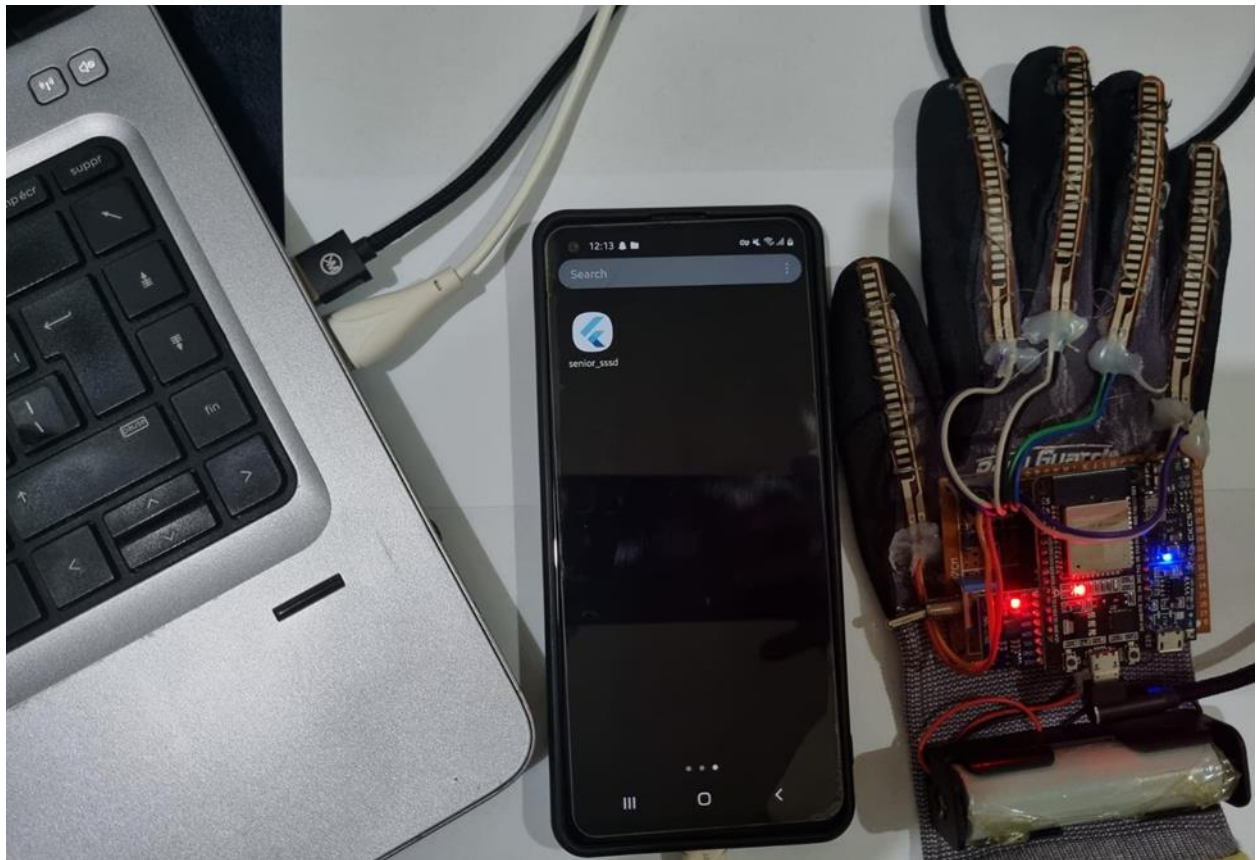


Figure 29: Hardware Results-4.

Software Results:

As illustrated in Fig. 30, the mobile application opens with a simple, user-friendly interface utilizing Flutter. The app shows the connection status with the glove via Bluetooth, and then indicates the gesture that was detected by a short piece of written text as well as audible speech. The design is simple, accessible, and clear to allow for clear communication for a user with hearing or speech.

Smart Speaking Signs Device

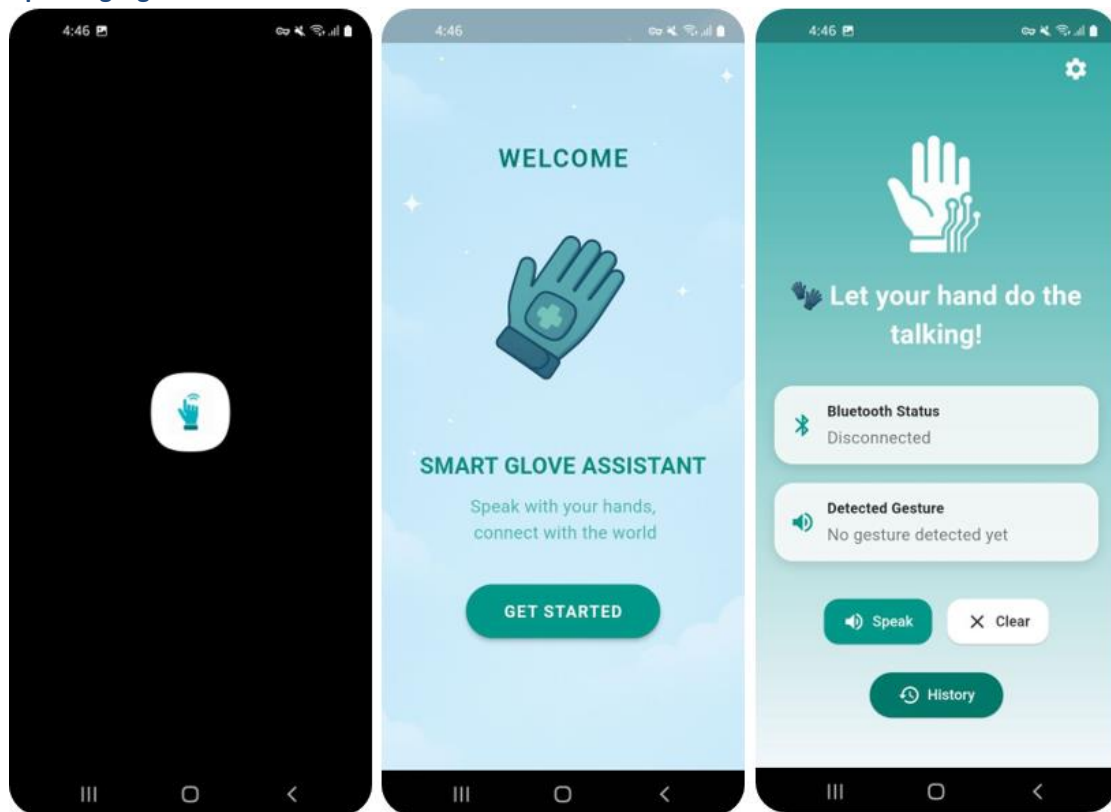


Figure 30: Software Results-1.

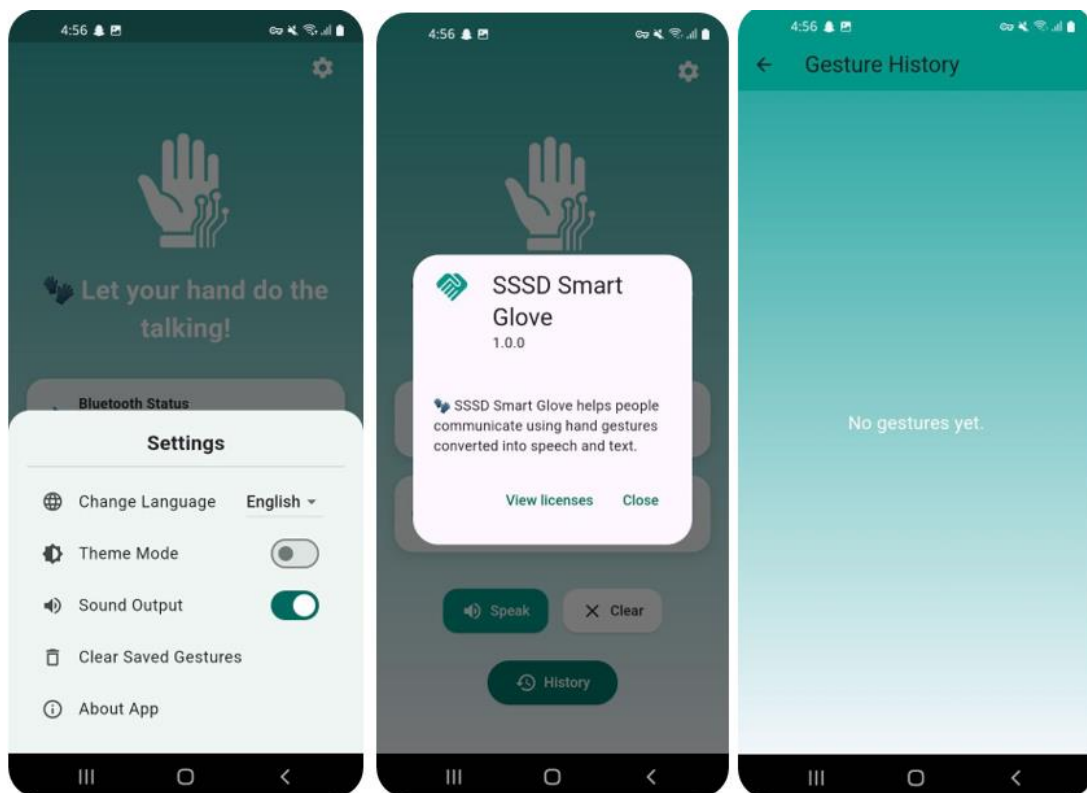


Figure 31: Software Results-2.

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On this page of the application, the user will be able to see the hand gesture recognition capability of the smart glove. After the ESP32 sends a gesture code via Bluetooth connection, the application will receive the code and will retrieve the matching phrase from the Firebase Realtime Database. The phrase that matches the gesture code will be displayed in Front of the user in a clear manner (See Fig. 32). Additionally, text-to-speech will be automatically triggered to speak the phrase to allow the user to communicate visually with the phrase shown on-screen, and audibly, with the speech.

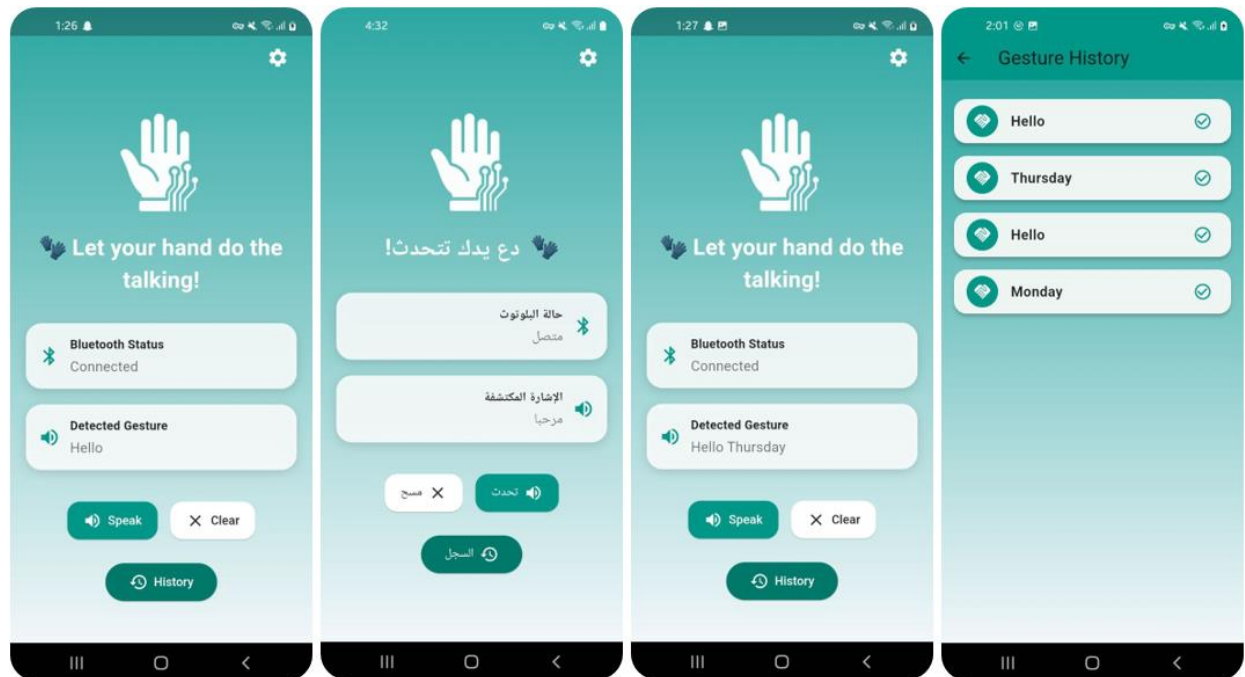


Figure 32: Software Results-3.

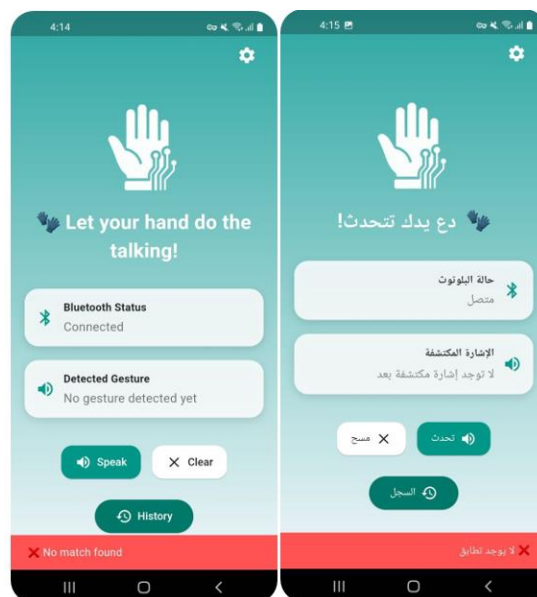


Figure 33: Software Results-4.

4.2 Discussions

During the first testing phase of the SSSD system hardware, we had the six hardware components assembled with 5 flex sensors and an MPU6050 motion sensor connected to the ESP32 microcontroller. We programmed the system to operate in real time, capturing gestures and converting them into corresponding gesture codes. These codes were then transmitted wirelessly to a mobile application via Bluetooth.

The mobile application, developed using Flutter, successfully received and recognized the gesture codes, matched them with the appropriate phrases stored in Firebase Realtime Database, and displayed the output text on screen along with voice using text-to-speech functionality.

A total of 40 predefined static hand gestures were implemented and tested. The system demonstrated an overall gesture recognition accuracy of approximately 87% under normal conditions, which is considered satisfactory for a prototype phase. The system responded in real time, with a processing and display delay of a few seconds for most gestures, ensuring seamless communication.

In terms of flexibility, the system allowed easy addition or modification of gesture entries in the database without requiring changes in hardware — making it adaptable for future updates, languages, or application domains (such as schools or public services).

Overall, the SSSD project successfully demonstrated that a sensor-based wearable device can provide a reliable, real-time, and adaptable solution to bridge communication barriers between deaf or mute individuals and their surrounding environment, especially in hospital settings where clarity and speed are essential.

CHAPTER 5: PROJECT MANAGEMENT

5.1 Tasks, Schedule and Milestones

Fig. 33 and 34 show the schedule reflecting the allocated time for completing each task according to the project priorities and time considerations, thus contributing to achieving the project goals within the specified deadlines.

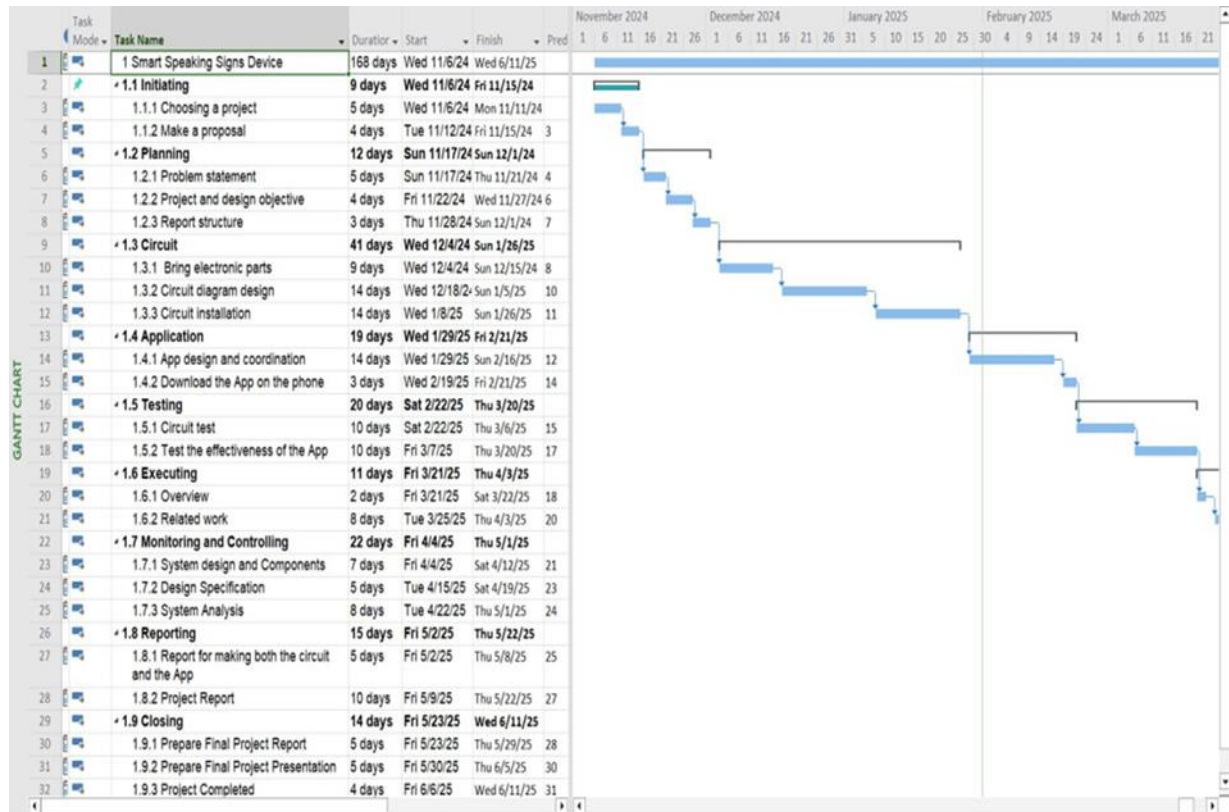


Figure 34: Gantt Chart-1.

Smart Speaking Signs Device

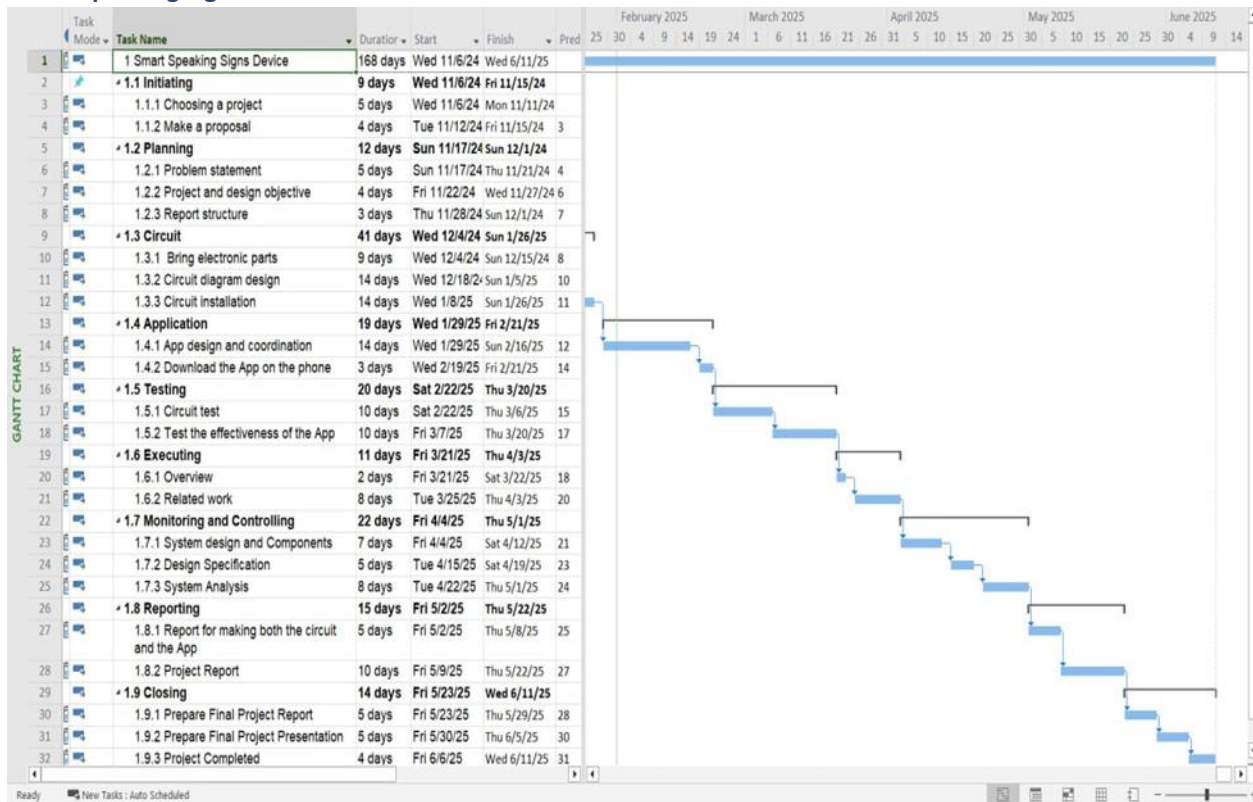


Figure 35: Gantt Chart-2.

5.2 Resources and Cost Management

The equipment and tools are used according to the specifications required to satisfy its function with minimum cost.

Table 3 shows the total cost of senior project items.

Component	Quantity	Estimated Cost	Total Cost
MPU6050 IMU Sensor	1	10 \$	10 \$
Flex Sensors	5	22 \$	110 \$
ESP32	1	13 \$	13 \$
Li-Ion Battery	1	3 \$	3 \$
Li-Ion Battery Charging Module	1	9 \$	9 \$
DC to DC Step-Up Booster Module	1	10 \$	10 \$
Wires & Connectors	-	3 \$	3 \$
Smart Glove (Material)	1	3 \$	3 \$
Total Estimated Cost	-	-	161 \$

Table 3: Resources and Cost Management

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When compared to similar assistive communication systems, the total cost of our project (161\$) is relatively low. For example, commercial smart gloves with gesture recognition capabilities typically range between 300–500 \$, and camera-based sign language recognition systems can exceed 700 \$ due to the cost of imaging hardware and processing units.

In contrast, our system achieves effective real-time gesture translation using low-cost components while maintaining portability, wireless communication, and ease of use, making it a more accessible and affordable solution, especially for healthcare and educational settings.

5.3 Lessons Learned

Throughout our time developing the SSSD system, we were able to experience many great lessons that positively influenced our time as well as our development. One of the major points was witnessing the impact teamwork can have when complemented with effective communication. With everyone immediately communicating together and knowing what role to take, we could collaborate well as a team and tackle issues creatively. One of the more interesting lessons and experience related to that was that time management was critical for us to be able to meet deadlines and maximize work efficiency. Clarifying goals and time objectives in our planning and organization created group accountability, clarity, and efficiency for each enjoyably, collectively completed phase of our project.

Researching and understanding projects of similar nature to ours in order to engage the project positively influenced one of the areas in learning we set out to achieve, which was sound decision making about our design work. Technically, we enhanced our skills while developing with fire animation training tools and Dart (which has SDKs /it's language for mobile development), and worked with Arduino and ESP32 (hardware). Throw in Firebase with depth of understanding, and use in mobile applications, we gained pragmatic understanding with utilizing real time data in applications.

Now, we can discuss project management all day! And project cost, which was essentially a 'good lesson', is a worthwhile inexpensive project management consideration; zero cost, the project continues indefinitely, but it should be considered, as it's a fundamental one in management projects or life.

CHAPTER 6: IMPACT OF THE ENGINEERING SOLUTION

6.1 Economical, Societal and Global

First, we will discuss the economic impact of the cost:

Our device is designed to be economically manageable to meet the needs of any user. The reasonable cost of access to a production prototype would provide better prospects for viable commercial production. As a result, the device is more replicable and therefore more accessible to people. This is useful in reducing costs, such as not hiring sign language interpreters when arranging a meeting with a public office or during medical visits.

Second, we will discuss the social impact of the product:

- The device is useful because it helps deaf and mute people communicate, thus improving and making their lives easier.
- In addition to helping those who cannot speak or hear, this project aims to benefit schools, hospitals and other organizations that help them.
- In some situations, the device eliminates the need for manual sign language interpretation, thus creating jobs and advancing assistive devices.

6.2 Environmental and Ethical

Finally, the ethical impact of the project:

The device is designed to comply with international environmental standards, and to be sustainable and environmentally friendly. Thanks to its efficient electronic components, it consumes a minimal amount of energy, which contributes to reducing greenhouse gas emissions worldwide. In addition, the glove has a very low environmental impact because it uses few natural resources, such as water or excess energy during its operation.

Instead of relying on rare or precious materials, the device is built using common and readily available components such as controllers, flexible sensors and communication modules. This strategy promotes ecological

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balance and reduces pressure on natural resources. Furthermore, the design itself allows for the recycling of components, which reduces the amount of electronic waste produced and encourages the reuse of materials.

In compliance with environmental regulations, the project ensures that manufacturing and waste disposal practices minimize harm to the environment. Sustainability is a key priority throughout the life cycle of the device, providing a practical solution to the deaf community while protecting the planet and complying with regulatory standards.

6.3 Other Issues

When designing the smart glove, we considered other factors in addition to sustainability. One was assurance that the Bluetooth communication is secure when sending gesture codes to the mobile app. We only transmit the gesture codes of a user's hand to the app and there is no identifiable or personal data, but we wanted to ensure a safe and stable connection.

Usability was also important – during development of the smart glove we made it lightweight and comfortable, especially for users with mobility limitations. As well, we created a glove that is easy to maintain by simply replacing the portions of the glove that may be damaged (e.g., sensors or wiring).

Lastly, we looked at the legality and regulation, especially for potential medical use. Following safety standards may help us ensure the smart glove can be utilized in actual medical environments.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Summary of Achievements of the Project Objectives

The SSSD project's primary result was to design a smart wearable system that translate the hand gestures used by individuals who have hearing and/or speech impairment into readable text and audible speech. The glove achieved this purpose utilizing both flex and MPU6050 sensors to work with the ESP32 controller.

The glove was successful in accommodating a considerable number of hand and finger movement which allowed the glove to capture and distinguish numerous gestures. The glove sent a code that represented the gesture variable via Bluetooth to the mobile app, the mobile app then matched the received code with a pre-saved list and retrieved the specific phrase from the Firebase Real Time Database. The matched phrase was displayed on the mobile app screen and generated into audible speech utilizing text to speech (TTS) as an output.

The system yielded accurate results with real-time response. The synchronous delivery of text and audio therefore provided uninterrupted and without disjointed communication allowing the SSSD project to fulfil the primary objectives of the project.

7.2 New Skills and Experiences Learnt

As part of the SSSD project, we interacted practically with flex sensors, the MPU6050, and the ESP32 microcontroller, which allowed us to Bluetooth communications, Arduino programming, and other systems develop supplementary Gesture-recognition systems.

Our mobile application development skills improved after learning Flutter and Dart because we integrated Firebase Realtime Database and added TTS functionalities.

Additionally, we further developed our ability to set team goals, plan and assign roles, devise and assess approaches, and monitor and report shifts and milestones—focusing on time-sensitive and multi-faceted deadlines essential for this project.

7.3 Recommendations for Future Work

Our ambition is to enhance our system by incorporating additional ML models that will detect more extensive gesture sets and accurately convert a variety of sign languages. Our system will be patterned on more elaborate and varied sets of data incorporating both regional and worldwide sign languages, powerfully integrating the ability to notice dynamic actions and nuanced gestures that display more in-depth connotation. Our project will also focus on enhancing the accuracy of gesture recognition, through the use of deep neural networks and techniques adopted from modern AI, to ensure that response times are rapid and error rates are minimized.

In addition, we would like to develop our mobile application by adding live support that allows users to contact translators or support staff at the same time, making it easier for them to interact instantly in different environments such as hospitals, schools, and government institutions. We also look forward to integrating voice assistant and AI technologies into the app so that we can make it even easier for users to have text and voice conversations.

In the future, we want to make the system multilingual, so that users can chat with them fluently without having to worry about changing languages. We will also support real-time translation in multiple languages so that the system can translate their gestures into text and voice in other languages and we can interact in multicultural environments. We will also customize the system for different types of wearable devices, including smart bracelets, smart glasses, and smart watches, thus increasing the flexibility of the device, making it easier for users to use the device according to their requirements. These improvements make the SSSD Social Sustainable Development System a global and integrated innovation that enables individuals with hearing and speech disabilities to interact freely and smoothly in different environments, thus improving their social integration and smooth communication with the community.

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	board?srsId=AfmBOoo4JcyXgTA4U2ioSWJHK_IKjZENCEfLaJ36FaJnFMeeJn9uK4Qg.
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APPENDICES

Appendix A:

Arduino IDE Code:

```
#include <Wire.h>
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>

#define MPU_addr 0x68
// إعداد BLE
BLEServer* pServer = nullptr;
BLECharacteristic* pCharacteristic = nullptr;
bool deviceConnected = false;

// UUIDs يجب أن تطابق ما في تطبيق Flutter
#define SERVICE_UUID      "4fafc201-1fb5-459e-8fcc-c5c9c331914b"
#define CHARACTERISTIC_UUID "beb5483e-36e1-4688-b7f5-ea07361b26a8"

bool imuAvailable = false;
const int flexPins[5] = {32, 33, 34, 35, 36};
int flexReadings[5];
int16_t accel[3];
int16_t gyro[3];

unsigned long lastReadTime = 0;
const unsigned long readInterval = 5000;
int readCounter = 0;
class MyServerCallbacks : public BLEServerCallbacks {
    void onConnect(BLEServer* pServer) {
```

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```
    deviceConnected = true;

    Serial.println("✔ Connected");
}

void onDisconnect(BLEServer* pServer) {
    deviceConnected = false;

    Serial.println("✗ Disconnected");
}

};

struct Gesture {
    String code;
    int flexMin[5];
    int flexMax[5];
    int16_t accelMin[3];
    int16_t accelMax[3];
    int16_t gyroMin[3];
    int16_t gyroMax[3];
};

Gesture gestures[] = {
    { "000001", // 1- شكراً
      {3145, 3057, 3195, 2755, 2791}, // flexMin[]
      {3262, 3071, 3203, 2763, 2985}, // flexMax[]
      {-4536, -9664, 12260},           // accelMin[]
      {-4188, -9508, 12424},           // accelMax[]
      {-215, 102, -65},                // gyroMin[]
      {-45, 218, 45}                   // gyroMax[]
    },
    { "000010", // 2- دكتور
      {3660, 2960, 3630, 3290, 2770}, // flexMin[] = original min - 50
      {3770, 3074, 4064, 3443, 2895}, // flexMax[] = original max + 50
      {4068, -14736, 5432},             // accelMin[] = original min - 200
      {4716, -13596, 7024},             // accelMax[] = original max + 200
    }
};
```

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```
{-407, 27, -436},          // gyroMin[] = original min - 100
{571, 408, 166}            // gyroMax[] = original max + 100
},
{ "000011", // 3- العين
  {3487, 2885, 3469, 3178, 3166}, // flexMin[] = min - 50
  {4009, 2996, 3764, 3357, 3379}, // flexMax[] = max + 50
  {1400, -16372, -780},           // accelMin[] = min - 200
  {1996, -15840, -188},           // accelMax[] = max + 200
  {-263, 76, -132},              // gyroMin[] = min - 100
  { 34, 317, 109}                // gyroMax[] = max + 100
},
{ "000100", // 4- الألم
  {3611, 3516, 3646, 3277, 3111}, // flexMin[] = min - 50
  {3740, 3671, 3851, 3399, 3217}, // flexMax[] = max + 50
  {1120, -15584, -5668},           // accelMin[] = min - 200
  {1740, -15092, -5064},           // accelMax[] = max + 200
  {-319, -10, -129},              // gyroMin[] = min - 100
  { 61, 373, 177}                // gyroMax[] = max + 100
},
{ "000101", // 5- كثير
  {3582, 2920, 3565, 3230, 2778}, // flexMin = min - 50
  {3787, 3042, 3937, 3439, 2919}, // flexMax = max + 50
  {712, -16144, 4156},             // accelMin = min - 200
  {2568, -14932, 5896},            // accelMax = max + 200
  {-412, -435, -820},              // gyroMin = min - 100
  {476, 704, 208}                 // gyroMax = max + 100
},
},
{ "000110", // 6- اليوم
  {3582, 2869, 3513, 3272, 2773}, // flexMin = min - 50
  {3719, 2999, 3938, 3407, 2947}, // flexMax = max + 50
  {1820, 12804, 9256},             // accelMin = min - 200
  {3100, 13528, 10356},            // accelMax = max + 200
  {-352, -59, -178},              // gyroMin = min - 100
  {122, 489, 179}                 // gyroMax = max + 100
}
```

```

},
{ "000111", // مكان 7-
  {3385, 3254, 3438, 2869, 2773}, // flexMin[] = min - 50
  {3572, 3385, 3554, 2991, 2900}, // flexMax[] = max + 50
  {276, 4668, 15404}, // accelMin[] = min - 200
  {1384, 5320, 16032}, // accelMax[] = max + 200
  {-365, 90, -126}, // gyroMin[] = min - 100
  { 92, 399, 178} // gyroMax[] = max + 100
},
{ "001000", // أين 8-
  {3123, 2926, 3342, 2884, 2814}, // flexMin = min - 50
  {3373, 3042, 3714, 3029, 3009}, // flexMax = max + 50
  {2868, -15012, -7800}, // accelMin = min - 200
  {5820, -12876, -6816}, // accelMax = max + 200
  {-644, -357, -983}, // gyroMin = min - 100
  {368, 649, 453} // gyroMax = max + 100
},
{ "001001", // لو سمحت 9-
  {3487, 2815, 3523, 3337, 2936}, // flexMin = min - 50
  {3716, 3010, 3715, 3573, 3169}, // flexMax = max + 50
  {1412, -14664, -9068}, // accelMin = min - 200
  {2816, -13324, -7328}, // accelMax = max + 200
  {-1152, -327, -510}, // gyroMin = min - 100
  {277, 750, 323} // gyroMax = max + 100
},
{ "001010", // أشعر 10-
  {2946, 2841, 2675, 2676, 2557}, // flexMin = min - 50
  {3083, 3024, 3361, 2816, 2928}, // flexMax = max + 50
  {11596, -7652, 7880}, // accelMin = min - 200
  {12748, -6460, 9232}, // accelMax = max + 200
  {-585, -257, -302}, // gyroMin = min - 100
  {368, 743, 255} // gyroMax = max + 100
},
{ "001011", // الدفع 11-
  {3180, 3091, 3533, 3295, 2638}, // flexMin = min - 50

```

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```
{3317, 3215, 3804, 3424, 2851}, // flexMax = max + 50
{2088, -7684, -15524}, // accelMin = min - 200
{3472, -6776, -14428}, // accelMax = max + 200
{-502, -31, -211}, // gyroMin = min - 100
{243, 580, 273} // gyroMax = max + 100
},
{ "001100", // 12- اناخير
{2955, 3204, 2951, 2650, 2782}, // flexMin[]
{3240, 3358, 4145, 2776, 2919}, // flexMax[]
{-2720, -16668, -4144}, // accelMin[]
{1620, -15444, 636}, // accelMax[]
{-471, -4059, -960}, // gyroMin[]
{1353, 365, 1117} // gyroMax[]
},
{ "001101", // 13- اريد التحدث مع الطبيب
{3650, 2680, 3400, 3380, 500},
{3750, 2780, 3550, 3450, 550},
{1160, -8100, 7600},
{1880, -7600, 8400},
{-4800, 100, -60},
{90, 270, 90}
},
{ "001110", // 14- مرحبا
{3085, 2940, 3263, 2743, 2814}, // flexMin[]
{3453, 3074, 3906, 3300, 3010}, // flexMax[]
{1052, -16300, -2704}, // accelMin[]
{1728, -15752, -1836}, // accelMax[]
{-287, 0, -190}, // gyroMin[]
{144, 506, 262} // gyroMax[]
},
{ "001111", // 15- عمل جيد
{3598, 3493, 3653, 3328, 2736}, // flexMin[]
{3801, 3648, 3837, 3479, 2859}, // flexMax[]
{13872, -8600, -168}, // accelMin[]
{15904, -6188, 2328}, // accelMax[]
```

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```
{-900, -1991, -1313},          // gyroMin[]
{3380, 1113, 1507}             // gyroMax[]
},
{ "010000", // 16- الم في البطن
  {2972, 2830, 3015, 2652, 2913},
{3108, 3029, 3241, 2786, 3117},
{16084, 292, -1416},
{16796, 1080, -52},
{-573, -15, -213},
{175, 377, 202}
},
{ "010001", // 17- صعوبة في التنفس
  {3193, 2986, 3045, 2318, 2716}, // flexMin[]
{3427, 3329, 3442, 2754, 2875}, // flexMax[]
{10032, -12348, 2588},           // accelMin[]
{10768, -11752, 3532},           // accelMax[]
{-318, -183, -173},             // gyroMin[]
{58, 420, 220}                  // gyroMax[]
},
{ "010010", // 18- أريد التكلم
  {2967, 2875, 2915, 2645, 2765}, // flexMin[]
{3093, 3031, 3253, 2767, 2925}, // flexMax[]
{-1192, -13580, 8432},           // accelMin[]
{340, -12208, 10768},            // accelMax[]
{-947, 96, -458},                // gyroMin[]
{1235, 484, 1055}               // gyroMax[]
},
{ "010011", // 19- سؤال
  {3580, 3198, 3603, 3244, 3115},
{3732, 3311, 3768, 3367, 3244},
{1000, -16444, 764},
{3404, -15516, 2108},
{-707, -422, -225},
{281, 971, 392}
},
```

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```
{ "010100", // أراك لاحقا-20
  {3511, 2860, 3505, 3225, 2771}, // flexMin[]
  {3708, 3028, 3847, 3449, 2916}, // flexMax[]
  {11212, -11184, 1948}, // accelMin[]
  {13236, -9240, 4208}, // accelMax[]
  {-1066, -75, -656}, // gyroMin[]
  {294, 529, 213} // gyroMax[]
},
{ "010101", // 21- #2
  {3021, 2881, 3454, 2907, 2968}, // flexMin[]
  {3279, 3013, 3844, 3121, 3121}, // flexMax[]
  {468, -16496, 980}, // accelMin[]
  {1576, -15512, 2292}, // accelMax[]
  {-880, -88, -207}, // gyroMin[]
  {212, 450, 545} // gyroMax[]
},
{ "010110", // 22- #3
  {3005, 2826, 3214, 3007, 2820}, // flexMin[]
  {3453, 2969, 3711, 3152, 2949}, // flexMax[]
  {-1728, -16292, -6980}, // accelMin[]
  {572, -14704, -2068}, // accelMax[]
  {-857, -186, -663}, // gyroMin[]
  {549, 500, 298} // gyroMax[]
},
{ "010111", // 23- #5
  {3646, 3116, 3582, 3111, 2846}, // flexMin[]
  {3936, 3328, 3935, 3312, 2996}, // flexMax[]
  {-2888, -16832, -4992}, // accelMin[]
  {2744, -14920, 1420}, // accelMax[]
  {-492, -8305, -2539}, // gyroMin[]
  {603, 554, 229} // gyroMax[]
},
{ "011000", // 24- #10
  {3662, 2862, 3523, 2692, 2928}, // flexMin[]
  {3842, 3041, 3927, 2848, 3069}, // flexMax[]
```


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```
{-3328, -16720, -4036},          // accelMin[]
{-1560, -14892, -764},           // accelMax[]
{-917, -330, -306},              // gyroMin[]
{644, 818, 473}                  // gyroMax[]
},
{ "011001", // 25- من
  {3552, 3500, 3383, 1973, 2880}, // flexMin[]
{3792, 3730, 3751, 2840, 3357},   // flexMax[]
{920, -18528, -9292},              // accelMin[]
{4752, -13108, -1044},             // accelMax[]
{-770, -640, -1101},              // gyroMin[]
{807, 12408, 6196}                 // gyroMax[]
},
{ "011010", // 26- مستشفى
{3021, 2942, 3470, 3216, 2820},   // flexMin[]
{3185, 3092, 3747, 3364, 2962},   // flexMax[]
{13708, -4868, 5448},              // accelMin[]
{15472, -3656, 6840},              // accelMax[]
{-323, -37, -81},                 // gyroMin[]
{66, 418, 254}                    // gyroMax[]
},
{ "011011", // 27- #6
  {2917, 2856, 3150, 3277, 2700}, // flexMin[]
{3192, 3039, 3444, 3497, 2979},   // flexMax[]
{400, -16476, -4952},              // accelMin[]
{2800, -15120, -2028},             // accelMax[]
{-722, -159, -205},               // gyroMin[]
{28, 511, 228}                    // gyroMax[]
},
{ "011100", // 28- #7
  {2945, 2897, 3589, 2677, 2796}, // flexMin[]
{3118, 3137, 4145, 2809, 3010},   // flexMax[]
{468, -16564, -3972},              // accelMin[]
{2600, -15220, -1856},             // accelMax[]
{-486, -134, -288},               // gyroMin[]
```

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```
{553, 583, 158}          // gyroMax[]
},
{ "011101", // 29- #8
  {3150, 3600, 3000, 3150, 400},
  {3300, 3750, 3200, 3250, 650},
  {1160, -8100, 7600},
  {1880, -7600, 8400},
  {-4800, 100, -60},
  {90, 270, 90}
},
{ "011110", // 30- مراقبة
  {3395, 4045, 3394, 2847, 2846},    // flexMin[]
  {3618, 4095, 3709, 3111, 3042},    // flexMax[]
  {-932, -16684, 896},                // accelMin[]
  {1708, -15324, 3092},                // accelMax[]
  {-535, -320, -423},                 // gyroMin[]
  {526, 829, 552}                     // gyroMax[]
},
{ "011111", // 31- الاعداد
  {3006, 4045, 3476, 2814, 3111},    // flexMin[]
  {3256, 4095, 3699, 2944, 3267},    // flexMax[]
  {-184, -16688, -4972},              // accelMin[]
  {4332, -15212, -320},               // accelMax[]
  {-1195, -162, -511},                // gyroMin[]
  {891, 1203, 1184}                   // gyroMax[]
},
{ "100000", // 32- اثنين
  {3157, 4045, 3245, 2703, 2980},    // flexMin[]
  {3320, 4095, 3489, 2848, 3369},    // flexMax[]
  {-1628, -16440, -5436},             // accelMin[]
  {1220, -14936, -3132},              // accelMax[]
  {-582, -804, -754},                 // gyroMin[]
  {427, 833, 584}                     // gyroMax[]
},
{ "100001", // 33- ثلاثة
```

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```
{3150, 3600, 3000, 3150, 400},
{3300, 3750, 3200, 3250, 650},
{1160, -8100, 7600},
{1880, -7600, 8400},
{-4800, 100, -60},
{90, 270, 90}
},
{ "100010", // خميس-34
  {3470, 3134, 3560, 3193, 2765}, // flexMin[]
{3654, 4145, 3697, 3330, 2900}, // flexMax[]
{2872, -16320, 4}, // accelMin[]
{7772, -13764, 3588}, // accelMax[]
{-1098, -1819, -824}, // gyroMin[]
{1085, 1736, 1452} // gyroMax[]
},
{ "100011", // قليل-35
  {3570, 4000, 3740, 3260, 2750}, // flexMin[]
{3690, 4120, 3850, 3375, 2950}, // flexMax[]
{1100, -16250, 700}, // accelMin[]
{1450, -14600, 1650}, // accelMax[]
{-1215, 150, -1350}, // gyroMin[]
{500, 700, 150} // gyroMax[]
},
{ "100100", // العصر-36
  {3000, 4000, 3140, 2735, 2800}, // flexMin[]
{3150, 4120, 3210, 2775, 2820}, // flexMax[]
{950, -16200, -2600}, // accelMin[]
{2700, -15800, 1000}, // accelMax[]
{-180, 120, -90}, // gyroMin[]
{600, 310, 340} // gyroMax[]
},
{ "100101", // استمر بالقوة-37
  {3450, 4080, 3400, 2680, 2850}, // flexMin[]
{3680, 4120, 3700, 2750, 2950}, // flexMax[]
```

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```
{-200, -9800, -13750},      // accelMin[]
{250, -9400, -13600},       // accelMax[]
{-350, 140, -90},           // gyroMin[]
{0, 330, 60}                // gyroMax[]
},
{ "100110", // ماذا تريد-38
  {3240, 3100, 3200, 2765, 2785}, // flexMin[]
{3310, 3190, 3430, 2820, 2920}, // flexMax[]
{4700, -4000, -16050},       // accelMin[]
{5500, -2200, -15500},       // accelMax[]
{-570, -50, -160},          // gyroMin[]
{770, 750, 420}              // gyroMax[]
},
{ "100111", // هذا رهيب-39
  {3215, 4080, 3180, 2650, 2730}, // flexMin[]
{3285, 4145, 3360, 2730, 2910}, // flexMax[]
{-3600, -15180, -6200},      // accelMin[]
{-2500, -14700, -5250},      // accelMax[]
{-540, -230, -600},          // gyroMin[]
{500, 520, 200}              // gyroMax[]
},
{ "101000", // محبة-40
  {3150, 3600, 3000, 3150, 400},
  {3300, 3750, 3200, 3250, 650},
  {1160, -8100, 7600},
  {1880, -7600, 8400},
  {-4800, 100, -60},
  {90, 270, 90}
}
};
```

```
void setup() {
  Serial.begin(115200);
```

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```
Wire.begin(21, 22);
Wire.beginTransmission(MPU_addr);
Wire.write(0x6B);
Wire.write(0);
if (Wire.endTransmission() == 0) {
    imuAvailable = true;
    Serial.println("✓ MPU6050 متصل بنجاح");
} else {
    Serial.println("⚠ MPU6050 غير متصل");
}

Serial.println("✔ الكود بدأ تشغيله بنجاح");

BLEDevice::init("SSSD");
pServer = BLEDevice::createServer();
pServer->setCallbacks(new MyServerCallbacks());

BLEService *pService = pServer->createService(SERVICE_UUID);
pCharacteristic = pService->createCharacteristic(
    CHARACTERISTIC_UUID,
    BLECharacteristic::PROPERTY_NOTIFY
);

pCharacteristic->addDescriptor(new BLE2902());
pService->start();
pServer->getAdvertising()->start();
Serial.println("🔊 BLE Advertising started...");
}

void loop() {
    if (millis() - lastReadTime >= readInterval) {
        lastReadTime = millis();
    }
}
```

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```
readCounter++;

if (readCounter >= 6) {
  Serial.println("تم أخذ 6 قراءات 📊");
  readCounter = 0; // إعادة العد
}

readSensors();

String code = detectGesture();

if (code != "") {
  Serial.println("تم التعرف على إشارة ✓: " + code);
} else {
  Serial.println("لا يوجد تطابق - سيتم إرسال 00000 ✖");
}

if (deviceConnected) {
  pCharacteristic->setValue(code != "" ? code.c_str() : "00000");
  pCharacteristic->notify();
  Serial.println("تم إرسال عبر BLE: " + String(code != "" ? code : "00000") + " 📶");
}

Serial.println("-----");
}
}

void readSensors() {
  Serial.println("---- Flex Sensors ----");
  for (int i = 0; i < 5; i++) {
    flexReadings[i] = analogRead(flexPins[i]);
    Serial.print("Flex["); Serial.print(i); Serial.print("] = ");
    Serial.println(flexReadings[i]);
  }
}
```

```

}

if (imuAvailable) {
  Serial.println("---- MPU6050 ----");
  Wire.beginTransmission(MPU_addr);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU_addr, 14, true);

  accel[0] = Wire.read() << 8 | Wire.read();
  accel[1] = Wire.read() << 8 | Wire.read();
  accel[2] = Wire.read() << 8 | Wire.read();
  Wire.read(); Wire.read(); // تجاهل الحرارة
  gyro[0] = Wire.read() << 8 | Wire.read();
  gyro[1] = Wire.read() << 8 | Wire.read();
  gyro[2] = Wire.read() << 8 | Wire.read();

  Serial.print("Accel: X="); Serial.print(accel[0]);
  Serial.print(" | Y="); Serial.print(accel[1]);
  Serial.print(" | Z="); Serial.println(accel[2]);

  Serial.print("Gyro: X="); Serial.print(gyro[0]);
  Serial.print(" | Y="); Serial.print(gyro[1]);
  Serial.print(" | Z="); Serial.println(gyro[2]);
}else {
  Serial.println("⚠ MPU6050 ");
}

Serial.println("-----");
}

String detectGesture() {
  for (int i = 0; i < sizeof(gestures) / sizeof(Gesture); i++) {
    bool flexMatch = true;

```

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```
for (int j = 0; j < 5; j++) {
    if (flexReadings[j] < gestures[i].flexMin[j] || flexReadings[j] > gestures[i].flexMax[j]) {
        flexMatch = false;
        break;
    }
}

if (flexMatch) {
    if (imuAvailable) {
        bool imuMatch = true;
        for (int j = 0; j < 3; j++) {
            if (accel[j] < gestures[i].accelMin[j] || accel[j] > gestures[i].accelMax[j])
                imuMatch = false;
            if (gyro[j] < gestures[i].gyroMin[j] || gyro[j] > gestures[i].gyroMax[j])
                imuMatch = false;
        }

        if (imuMatch) {
            Serial.println("★ تطابق عالي الدقة (IMU).");
        } else {
            Serial.println("△ تطابق بالفليكس فقط (IMU).");
        }
    }

    return gestures[i].code;
}

return ""; // لا يوجد تطابق
}
```